

WHAT MATTERS FOR REPRESENTATION ALIGNMENT: GLOBAL INFORMATION OR SPATIAL STRUCTURE?

Anonymous authors
Anonymous submission

ABSTRACT

Representation alignment (REPA) regularizes generative training by distilling the representations of a strong, pretrained vision encoder into the intermediate features of a diffusion transformer. We study a precise question about the mechanism underlying this regularization: which property of the target representation governs its downstream generation quality, its *global* semantic content (operationalized as ImageNet-1K linear-probing accuracy) or its spatial structure (operationalized as the pairwise cosine similarity between patch tokens)? The prevailing hypothesis holds that a target representation with stronger global semantic performance yields better generation. To test this hypothesis, we conduct a large-scale empirical analysis spanning 27 distinct vision encoders and three model scales. The results contradict the prevailing view: it is spatial structure, rather than global performance, that governs the generation quality obtained from a target representation. Motivated by this finding, we introduce two minimal modifications to the REPA recipe that specifically amplify the transfer of *spatial* information. Concretely, we substitute the standard MLP projection layer in REPA with a single convolutional layer and prepend a spatial normalization layer applied to the external representation. The resulting method, termed iREPA and expressible in fewer than four lines of code, consistently accelerates the convergence of REPA across a diverse set of vision encoders, model sizes, and training variants (REPA, REPA-E, MeanFlow, JiT). These results motivate a reexamination of the operating mechanism of representation alignment and of how it can be exploited for more efficient training of generative models.

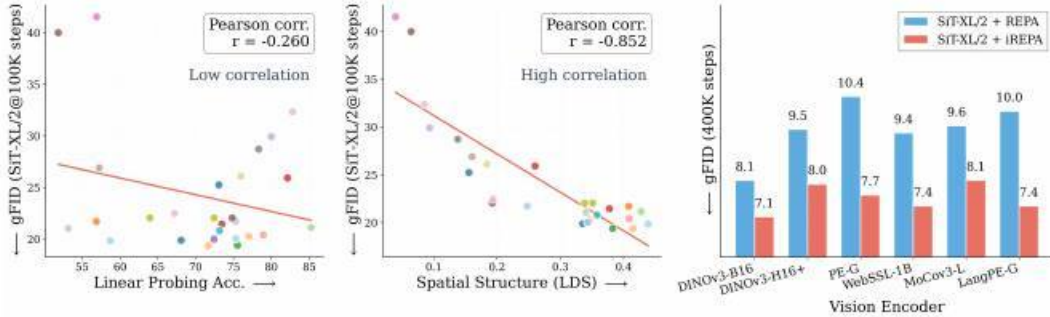


Figure 1: **What matters for representation alignment?** *Left:* Correlation analysis over 27 diverse vision encoders. Contrary to the prevailing assumption, the generation performance of a target representation is governed by its spatial structure rather than by its global performance (linear-probing accuracy). *Right:* Motivated by this finding, we introduce two minimal modifications that amplify the transfer of spatial features from the target representation to the diffusion model; the resulting method consistently accelerates the convergence of REPA across diverse settings.

1 INTRODUCTION

Representation alignment has emerged as an effective mechanism for accelerating the training of diffusion transformers (Ma et al., 2024; Peebles & Xie, 2023). By constraining the internal



Figure 2: **Motivating examples — spatial structure matters.** Comparison of encoder metrics exhibiting an inverse relationship between ImageNet accuracy and generation quality. *Left:* PE-G, despite substantially higher validation accuracy (82.8% vs. 53.1%), yields worse generation than SpatialPE-B (Bolya et al., 2025). *Right:* Analogously, WebSSL-1B (Fan et al., 2025) attains substantially higher global performance (76.0% vs. 53.1%) yet worse generation. **Spatial Self-Similarity:** Spatial structure provides a more reliable predictor of generation quality than global performance; see §3 for the spatial structure metric. All results reported at 100K iterations using SiT-XL/2 and REPA.

representations of a diffusion model to align with those of a pretrained self-supervised visual encoder, recent methods attain substantial gains in both convergence speed and final generation quality (Yu et al., 2024; Leng et al., 2025a). Despite this empirical evidence, the precise mechanism by which self-supervised features improve diffusion training remains poorly characterized. A specific question therefore persists: is the observed improvement attributable primarily to the transfer of richer global semantic content, as quantified by linear-probing accuracy, or to the transfer of spatial structure, as characterized by the pairwise relationships among patch-token representations?

Characterizing this mechanism is central to advancing generative training, since it determines how one should select a target representation and how its benefits can be maximized. The prevailing assumption is that an encoder’s utility for representation alignment is strongly correlated with its ImageNet-1K validation accuracy, taken as a proxy for global semantic understanding (Oquab et al., 2024; Chen et al., 2021). Under this assumption, target representations with higher ImageNet accuracy are hypothesized to induce better generation (Yu et al., 2024). Consistent with this view, increases in the linear-probing accuracy of diffusion features are routinely reported as evidence for the effectiveness of representation alignment, further entrenching global semantic content as the presumed primary driver of the improvement. The following statement from REPA (Yu et al., 2024) summarizes this prevailing understanding:

“When a diffusion transformer is aligned with a pretrained encoder that offers more semantically meaningful representations (i.e., better linear probing results), the model not only captures better semantics but also exhibits enhanced generation performance, as reflected by improved validation accuracy with linear probing and lower FID scores.”

In this work, we subject this conventional wisdom to a systematic test, isolating whether global semantic content or spatial structure is the operative factor in representation alignment. Through a large-scale empirical analysis over a diverse set of vision encoders, including recent large-scale vision foundation models such as WebSSL (Fan et al., 2025), DINOv3 (Siméoni et al., 2025), perception encoders (Bolya et al., 2025), and C-RADIO (Heinrich et al., 2024), we establish three findings that run counter to the prevailing assumption.

Higher validation accuracy does not imply a better target representation for generation. Contrary to the prevailing assumption, vision encoders with higher global semantic performance, as measured by ImageNet-1K linear-probing accuracy, frequently yield inferior generation. As a concrete instance, consider PE-Spatial-B, a spatially tuned encoder distilled from PE-Core-G (Bolya et al., 2025). Although PE-Spatial-B attains substantially lower patch-token linear-probing accuracy¹ than PE-Core-G (53.1% versus 82.8%), it produces better generation under REPA (Figure 2). An analogous inversion holds for WebSSL-1B (Fan et al., 2025), which exhibits markedly higher global performance (76.0% versus 53.1%) yet worse generation. In §2 we show that this pattern recurs across a broad range of target representations, indicating a systematic principle governing how representation alignment benefits diffusion training.

¹Following REPA (Yu et al., 2024), we quantify the global semantic performance of an external representation via linear-probing accuracy on its patch tokens, since only patch tokens participate in representation alignment.

Spatial structure, not global information, determines generation performance. To make this claim quantitative, we define several tractable metrics that measure the spatial self-similarity structure (Shechtman & Irani, 2007) of patch tokens (§3). We then conduct a large-scale correlation analysis relating REPA generation FID to linear-probing accuracy and to spatial structure, across 27 vision encoders and 3 model sizes (SiT-B, SiT-L, SiT-XL). Every spatial structure metric attains a substantially stronger association with generation FID (Pearson $|r| > 0.852$) than ImageNet-1K validation accuracy ($|r| = 0.26$), which carries almost no predictive power. Two additional experiments corroborate this conclusion by demonstrating that target representations carrying very little global information nonetheless yield large REPA gains. First, SAM2 produces better generation under REPA than encoders whose ImageNet accuracy is roughly 60% higher (Fig. 3). Second, classical, purely spatial descriptors such as SIFT (Lowe, 1999) and HOG (Dalal & Triggs, 2005), while not optimal, still deliver appreciable gains under representation alignment (§3).

Amplifying spatial features improves convergence. Building on the preceding analysis, we introduce two minimal modifications (fewer than four lines of code) that specifically amplify the transfer of spatial information from the target representation to the diffusion model. (1) We insert a spatial normalization layer that increases the spatial contrast of the target representation. (2) We show that the standard MLP projection layer, used to map diffusion features onto the target-representation dimensionality, attenuates spatial information (Figure 6a), and we replace it with a single convolutional layer that preserves this structure. The resulting method, iREPA, consistently accelerates the convergence of REPA across variation in encoders, model sizes, and training recipes, including REPA-E (Leng et al., 2025a) and MeanFlow (Geng et al., 2025) with REPA.

The principal contributions of this work are as follows:

- We conduct a large-scale empirical analysis establishing that spatial structure, rather than global semantic content, is the determining factor in the effectiveness of representation alignment.
- We define a simple, inexpensively computable Spatial Structure Metric (SSM) whose correlation with downstream FID is substantially higher than that of linear-probing accuracy.
- We propose two minimal modifications that amplify the transfer of spatial information from the target representation to the diffusion features. The resulting method yields consistent convergence-speed improvements over REPA across variation in target representation, model architecture, and training recipe (REPA, REPA-E, MeanFlow with REPA, and JiT with REPA).

2 MOTIVATION: GLOBAL INFORMATION MATTERS LESS

We begin with a set of motivating observations demonstrating that higher global performance of a target representation (ImageNet-1K accuracy) does not entail better generation under REPA. In §3 we show that these observations, which lack a satisfactory explanation under the global-information hypothesis, are accounted for by the spatial structure of the target representation.

Recent vision encoders. We examine several recent vision encoders, including Perception Encoders (Bolya et al., 2025), WebSSL (Fan et al., 2025), and DINOv3 (Siméoni et al., 2025). Consider PE-Spatial-B (80M), a compact spatially tuned encoder distilled from PE-Core-G (1.88B) (Bolya et al., 2025). As reported in Figure 2, PE-Core-G attains substantially higher ImageNet-1K accuracy (82.8% versus 53.1%) yet, when used as the target representation for REPA, yields worse generation (FID 32.3 versus 21.0). The same inversion holds for WebSSL-1B (1.2B): despite substantially higher ImageNet-1K accuracy (76.0% versus 53.1%), it produces worse generation as a REPA target (FID 26.1 versus 21.0).

SAM2 outperforms vision encoders with much higher ImageNet-1K accuracy. To probe the effect of minimal global information on generation, we analyze the SAM2-S vision encoder (46M) (Ravi et al., 2024), a compact model carrying very little global information, with a validation accuracy of only 24.1% (Fig. 3a). Notwithstanding this low accuracy, when used as a REPA target SAM2-S attains a lower (better) FID than vision encoders with markedly higher ImageNet-1K accuracy, *e.g.*, PE-Core-G (82.8%).

Larger models within the same encoder family can yield similar or worse generation performance. It is commonly assumed that, within a fixed encoder family, larger models learn better representations, as measured by ImageNet-1K accuracy. For representation alignment, however, larger variants often produce comparable (DINOv2) or even *worse* (PE, C-RADIO) generation (Fig. 3b).

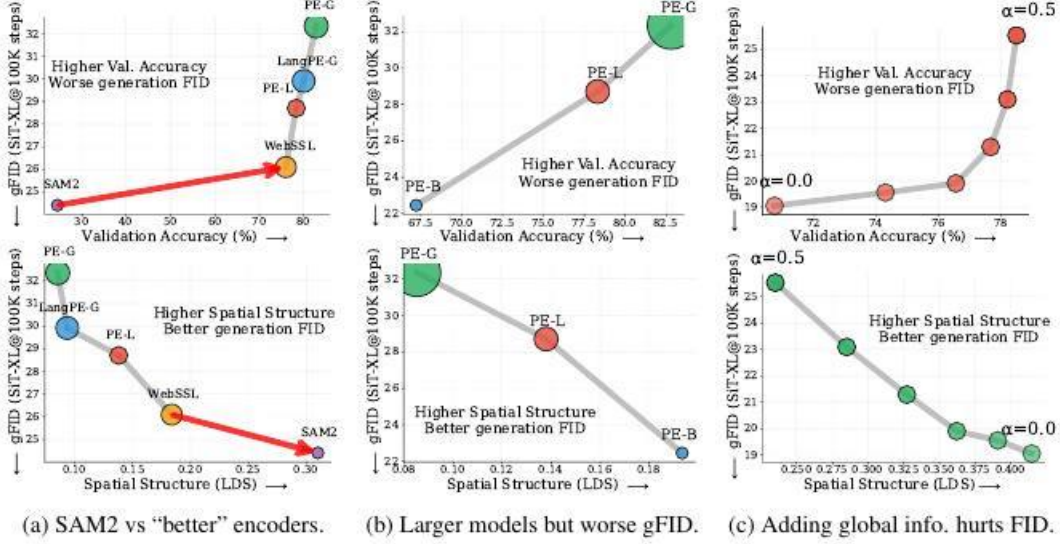


Figure 3: **Higher global information does not imply better REPA performance.** **Top row:** Three trends demonstrating that global performance is not predictive of generation FID under REPA. (a) SAM2-S, with a validation accuracy of only 24.7%, yields better REPA generation than encoders whose validation accuracy is roughly 60% higher. (b) Larger encoders within a family attain higher validation accuracy yet worse generation. (c) Injecting global information into patch tokens via the CLS token raises global performance but degrades generation. **Bottom row:** Spatial structure, rather than global performance, provides a reliable predictor of generation; see §3 for the detailed large-scale analysis across spatial structure metrics. All results are reported w/o classifier free guidance, SiT-XL/2 w/ REPA and 250 NFE (Yu et al., 2024) for inference.

Yu et al. (2024) report a similar observation for DINOv2 and attribute it to the hypothesis that “all DINOv2 models being distilled from the DINOv2-g model and thus sharing similar representations”. We subsequently show that this trend admits a more precise explanation in terms of spatial structure (§3).

Injecting additional global information degrades generation. To test directly whether additional global information benefits representation alignment, we conduct a controlled experiment that injects global semantics into the local patch tokens via the CLS token, using DINOv2 as the encoder. The mixing operation is $\mathbf{p}_i^{\text{new}} = \mathbf{p}_i + \alpha \cdot \mathbf{c}$, where $\mathbf{p}_i$ is the i^{th} patch token, $\mathbf{c}$ is the CLS token, and $\alpha \in [0, 0.5]$ is a scalar controlling the mixing strength. As α increases from 0 to 0.5 (Figure 3c), linear-probing accuracy rises monotonically from 70.7% to 78.5%, whereas generation quality degrades substantially, with FID rising from 19.2 at $\alpha = 0$ to 25.4 at $\alpha = 0.5$. Thus, increasing global semantic content while holding all else fixed is detrimental to generation.

Taken together, these observations establish that the global performance of a representation is not a reliable predictor of its REPA effectiveness. In §3 we show that spatial structure provides a substantially more informative predictor.

- **Finding 1.** Better global semantic information (e.g., ImageNet-1K accuracy) does not imply a better representation for generation.

3 SPATIAL STRUCTURE MATTERS MORE

We hypothesize that the generation quality obtained from a target representation is governed by its spatial structure rather than by its global information. To render this hypothesis testable, we first define several tractable metrics that quantify the spatial self-similarity structure (Shechtman & Irani, 2007) of a target representation. We then demonstrate that every such metric (i) correlates far more strongly with gFID than ImageNet-1K accuracy does, and (ii) accounts for the observations in §2 that the global-information hypothesis fails to explain.

Measuring spatial self-similarity structure. Given an image $\mathcal{I}$ and a vision encoder $\mathcal{E}$, we define the following objects:

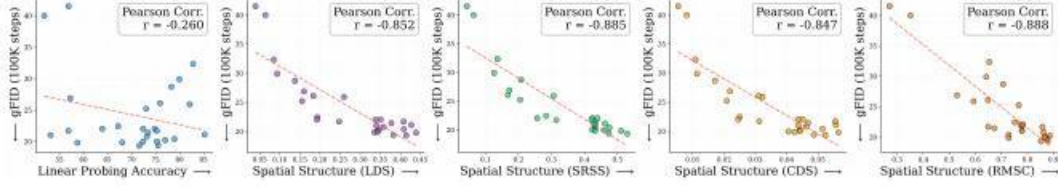


Figure 4: Spatial structure correlates more strongly with generation quality than linear probing. Correlation analysis over 27 diverse vision encoders, using SiT-XL/2 and REPA. Linear probing is weakly correlated with FID (Pearson $|r| = 0.260$), whereas every spatial structure metric, LDS ($|r| = 0.852$), SRSS ($|r| = 0.885$), CDS ($|r| = 0.847$), and RMSC ($|r| = 0.888$), attains a substantially stronger correlation with generation performance. See Fig. 10 for per-encoder labeled plots.

- *Self-similarity.* Let $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ denote the extracted patch representations over $T = H \times W$ patches. Let the kernel $K_{\mathcal{X}}(\cdot, \cdot) \in \mathbb{R}$ quantify appearance self-similarity (Shechtman & Irani, 2007) between pairs of patch tokens; we instantiate it as the cosine kernel $K_{\mathcal{X}}(t, t') = \frac{\langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle}{\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2}$.
- *Spatial distance.* Let $\mathcal{P} \in \mathbb{N}^{T \times 2}$ collect the lattice coordinates of the T tokens, and let $d(\cdot, \cdot) \in \mathbb{N}$ denote the Manhattan distance between a pair of tokens in this coordinate space.

We define a *spatial* self-similarity metric as a summary of how the self-similarity $K_{\mathcal{X}}$ varies with the lattice distance d between patch tokens. Larger values indicate stronger spatial organization, i.e., nearby patches are more similar to one another than distant patches. As the default instantiation we adopt a correlogram-contrast (local versus distant) metric (Huang et al., 1997):

$$\text{LDS}(\mathcal{X}, \mathcal{P}) = \mathbb{E}_{d(t, t') \in (0, r_{\text{near}})} K_{\mathcal{X}}(t, t') - \mathbb{E}_{d(t, t') \geq r_{\text{far}}} K_{\mathcal{X}}(t, t').$$

The final spatial self-similarity score (LDS) is obtained as the expectation of this contrast over patch representations $\mathcal{X}$. We set $r_{\text{near}} = r_{\text{far}} = H/2$; the resulting correlations are empirically insensitive to the precise values of these radii. Appendix B provides the full definitions together with alternative spatial metrics (CDS, SRSS, RMSC) that are similarly effective (see Fig. 4).

Spatial structure correlates far more strongly with generation performance than linear probing.

We conduct a large-scale correlation analysis over 27 diverse vision encoders. As reported in Figure 4, the conventionally used linear-probing accuracy exhibits negligible correlation with generation FID (Pearson $|r| = 0.26$), whereas every SSM metric attains a much stronger correlation (Pearson $|r| > 0.85$).

Generalization across model scales. We verify that this relationship persists across model scales (SiT-B, SiT-L, SiT-XL) in Figure 5. Across all scales, linear probing remains weakly correlated with generation performance ($|r| < 0.306$), while spatial structure maintains a much stronger correlation ($|r| > 0.826$).

- **Finding 2.** *Spatial structure correlates much higher with generation performance than linear probing. We next use spatial metrics to explain previously unexplained trends.*

Spatial structure accounts for previously unexplained trends. As established in §2, global performance (ImageNet-1K accuracy) is not predictive of representation-alignment effectiveness. We now show that the spatial metrics defined above supply the missing predictor, resolving each of the four observations in turn.

(1) *PE-Spatial-B versus PE-Core-G:* Figure 2 shows that PE-Core-G attains substantially higher ImageNet-1K accuracy (82.8% versus 53.1%) yet worse REPA generation (FID 32.3 versus 22.0). The spatial structure metric resolves the apparent paradox: despite its lower global performance, PE-Spatial-B exhibits markedly stronger spatial structure than PE-Core-G, consistent with its superior generation.

(2) *SAM2 outperforms “better” vision encoders:* §2 shows that although SAM2 attains far lower ImageNet-1K accuracy (24.1%), it yields better generation than encoders with roughly 60% higher accuracy. As shown in Fig. 3a, this gain is attributable to SAM2’s stronger spatial structure.

(3) *Larger models within an encoder family underperform:* As shown in Fig. 3b, larger models within a family attain higher ImageNet-1K accuracy but can exhibit weaker spatial structure, which explains

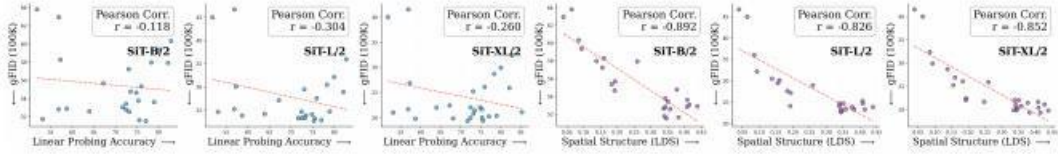
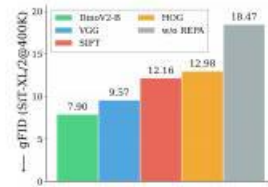


Figure 5: **Correlation analysis across model scales.** At every model scale, spatial structure (right) attains a consistently higher correlation with gFID than linear probing (left).

their worse REPA generation. This is consistent with Yu et al. (2024), who observe that larger models can yield comparable or worse generation.

(4) *Injecting global information into patch tokens via the CLS token degrades generation:* In Figure 3c, mixing the CLS token into patch tokens raises global performance but reduces the spatial contrast among patch tokens, inducing spurious high similarity between otherwise unrelated tokens (e.g., foreground object and background). The attendant loss of spatial structure explains the observed degradation in generation.

If spatial structure is the operative factor, can SIFT or HOG features serve as REPA targets? They can. While not optimal, the classical spatial descriptors SIFT (Lowe, 1999) and HOG (Dalal & Triggs, 2005), as well as intermediate VGG features (Simonyan & Zisserman, 2014), each yield measurable gains under REPA. This provides direct evidence that representation alignment can benefit from purely spatial features, without any additional global information.



Do the spatial metrics also account for the gains obtained with REPA?

They do. As shown, the proposed spatial metrics explain both the gains of standard REPA and those of our improved training recipe (iREPA), introduced next in §4.



4 iREPA: IMPROVING REPRESENTATION ALIGNMENT BY ACCENTUATING WHAT MATTERS

To further isolate the role of spatial structure in representation alignment, we introduce two minimal modifications to the original REPA training recipe, each designed to enhance the transfer of spatial features from the teacher (vision encoder) to the student (diffusion transformer).

Convolutional projection layer in place of the MLP. Standard REPA employs a three-layer MLP projection to map the diffusion feature dimensionality onto that of the external representation. As shown in Figure 6a, this projection is lossy: it attenuates the spatial contrast between patch tokens. We therefore replace the MLP with a lightweight convolutional layer (kernel size 3, padding 1) applied directly over the spatial grid. Owing to its locality-preserving inductive bias, the convolution retains local spatial relationships that the MLP discards.

Spatial normalization layer. In agreement with Siméoni et al. (2025), we find that the patch tokens of pretrained vision encoders contain a substantial global component, evidenced by the high linear-probing accuracy attained by the mean of the patch tokens (Figure 14). Moreover, Figure 6b shows that mixing this global component (the mean of the patch tokens) into the individual patch tokens raises global performance but simultaneously reduces their spatial contrast, so that a token (e.g., on a foreground object) exhibits spurious high similarity with unrelated tokens (e.g., on the background).

Given the analysis in §3, we hypothesize that discarding this global component in exchange for higher spatial contrast among patch tokens yields a stronger spatial signal (the pairwise similarities among patch tokens) and hence improves REPA. Concretely, we prepend a spatial normalization layer (Ulyanov et al., 2016) to the patch tokens of the target representation:

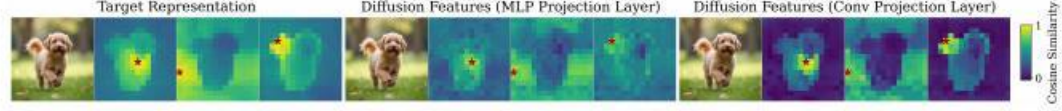
$$\mathbf{y} = \frac{\mathbf{x} - \gamma \mathbb{E}[\mathbf{x}]}{\sqrt{\text{Var}[\mathbf{x}] + \epsilon}},$$

where $\mathbf{x} \in \mathbb{R}^{B \times T \times D}$ denotes the patch tokens, the expectation and variance are taken over the spatial dimension, γ scales the removed global (mean) component, and $\epsilon = 10^{-6}$ ensures numerical stability.

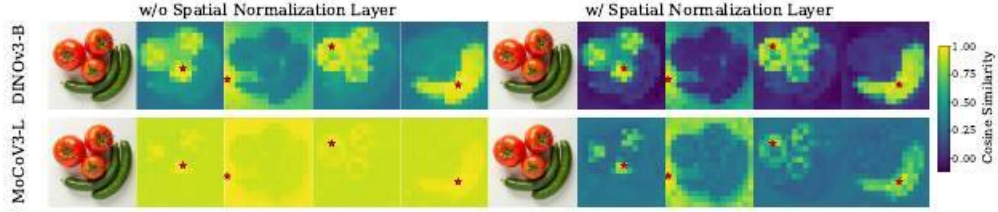
Algorithm 1 Summary of key iREPA modifications to standard REPA training

```
# 1. Conv projection instead of MLP
proj_layer = nn.Conv2d(D_in, D_out, kernel_size=3, padding=1)
```

```
# 2. Spatial normalization on encoder features [B, T, D]
x = x - gamma * x.mean(dim=1, keepdim=True)
x = x / (x.std(dim=1, keepdim=True) + 1e-6)
```



(a) *Simpler projection layer for REPA.* The standard MLP projection layer in REPA (middle) discards spatial information when mapping the target representation (left) onto the diffusion features; replacing it with a single convolutional layer preserves this spatial information (right).



(b) *Spatial normalization layer.* The patch tokens of pretrained vision encoders contain a global component that suppresses spatial contrast, inducing appreciable cosine similarity between tokens in one semantic region (e.g., tomato) and unrelated tokens (e.g., background or cucumber). We hypothesize that discarding this global component in exchange for higher spatial contrast among patch tokens improves generation.



(c) *Overall effect of the improved training recipe (iREPA) on the spatial structure of the diffusion features.*

Figure 6: Motivation for the two minimal modifications (iREPA) that enhance the transfer of spatial features from the target representation to the diffusion features. All results reported using SiT-L/2 with REPA at 100K iterations.

4.1 EXPERIMENTS

In this section, we empirically validate the improved training recipe through an extensive evaluation on ImageNet 256×256 . Specifically, we address the following research questions:

- Does iREPA consistently accelerate the convergence of diffusion transformers relative to REPA across a diverse set of external representations? (Figure 7, 8, 12, 13, 15)
- Does iREPA scale with model size and generalize across variations in the training configuration? (Table 1 [a,b,c], 2, 4, 5, 6, and Figure 12, 16)
- Does iREPA generalize to more recent representation-alignment methods, such as REPA-E (Leng et al., 2025a) and MeanFlow with REPA (Geng et al., 2025), and to pixel-space diffusion models such as JiT with REPA (Li & He, 2025b)? (Table 3 [a,b])

Convergence Speed. We evaluate the convergence behavior of iREPA across a range of vision encoders (DINOv3-B, WebSSL-1B, PE-Core-G, CLIP-L, MoCov3, PE-Lang-G) and two model sizes (SiT-XL/2 and SiT-B/2), with results reported in Fig. 7. Across every combination of target representation and model size, iREPA attains faster convergence than the REPA baseline.

Target representation. We assess the generalization of iREPA across vision encoders in Fig. 8 and Table 4. iREPA improves generation quality for every encoder considered. Comprehensive comparisons spanning all 27 encoders are reported in Appendix C, E.

Encoder size. We evaluate the generalization of iREPA across encoder scales, PE-B (90M), PE-L (320M), and PE-G (1.88B), in Table 1a. iREPA improves performance at every encoder scale.

Enc. Size	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Re. $\uparrow$
PE-B (90M)	59.6	22.5	6.23	0.63	0.60
+iREPA	73.3	17.5	6.04	0.66	0.62
PE-L (320M)	48.6	28.7	6.53	0.59	0.60
+iREPA	76.3	17.6	6.36	0.65	0.61
PE-G (1.88B)	42.7	32.3	6.70	0.57	0.59
+iREPA	70.8	19.5	6.35	0.64	0.61

(a) Vision Encoder size

Model Size	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Re. $\uparrow$
SiT-B	27.5	49.50	7.00	0.46	0.59
+iREPA	34.1	43.37	7.87	0.50	0.60
SiT-L	55.7	24.10	6.25	0.62	0.60
+iREPA	66.9	20.28	6.14	0.63	0.62
SiT-XL	70.3	19.06	5.83	0.65	0.61
+iREPA	77.9	16.96	6.26	0.66	0.61

(b) Model size

Aln. Depth	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Re. $\uparrow$
Layer 4	28.1	49.0	6.83	0.46	0.60
+iREPA	41.7	36.0	6.61	0.52	0.62
Layer 6	26.9	50.6	7.00	0.46	0.59
+iREPA	42.4	35.3	6.90	0.53	0.62
Layer 8	24.7	54.8	7.35	0.44	0.58
+iREPA	38.3	38.9	7.19	0.51	0.62

(c) Encoder depth (SiT-B/2)

Table 1: **Variation in training configuration.** iREPA yields consistent gains over the REPA baseline across diverse training configurations. Unless otherwise specified, all results use DINOv2 as the encoder, SiT-XL/2, 100K steps, and vanilla REPA as the baseline.

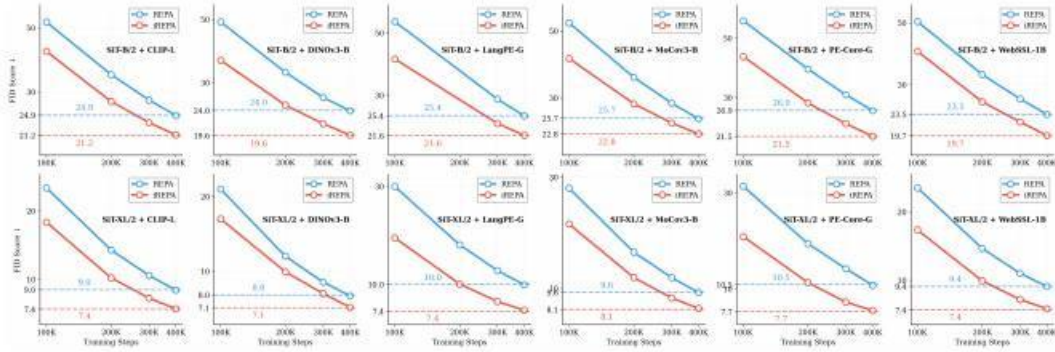


Figure 7: **Amplifying spatial features consistently accelerates convergence.**

Notably, the relative improvement grows monotonically with encoder size (22.2% for PE-B, 38.8% for PE-L, and 39.6% for PE-G).

Scalability. We evaluate the scalability of iREPA across diffusion-model scales in Table 1b. The spatial modifications improve performance at every scale, and the relative gain increases with model size, establishing that the spatial improvements scale favorably with model capacity.

Encoder depth. Table 1c evaluates the generalization of iREPA across alignment depths, with all results obtained using SiT-B/2 at 100K iterations and DINOv3-B as the target representation. iREPA improves over the REPA baseline at every alignment depth.

Abalation on different components. We ablate the two components of iREPA in Table 2. Each component in isolation, the spatial normalization layer and the convolutional projection layer, significantly improves generation quality over the REPA baseline, and their combination yields the best results.

Training recipe. Finally, we evaluate the generalization of iREPA across training recipes, specifically REPA-E (Leng et al., 2025a) and MeanFlow with REPA (Geng et al., 2025). Table 3 shows that the spatial modifications of iREPA yield convergence gains under both recipes.

Classifier-free guidance. Table 4 reports the generation quality of iREPA under classifier-free guidance (CFG) across vision encoders at 400K training iterations. For every encoder, iREPA attains faster convergence both with and without CFG.

Pixel-space diffusion (JiT). We further evaluate iREPA on the pixel-space diffusion model JiT-B (Li & He, 2025b). Figure 9 compares REPA and iREPA under JiT (Li & He, 2025b). Amplifying the transfer of spatial information consistently accelerates convergence across vision encoders (e.g., DINOv2, DINOv3, PE).

- **Finding 3.** Accentuating transfer of spatial structure from target representation to diffusion features helps boost convergence speed with representation alignment (REPA).

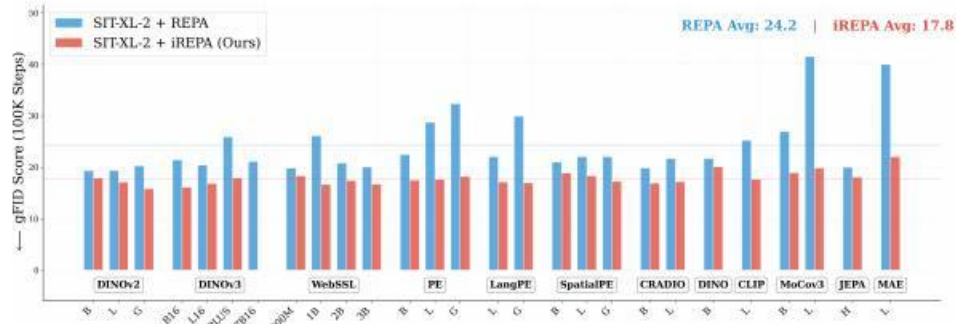


Figure 8: **Variation in target representation.** Across all 27 vision encoders, amplifying the transfer of spatial features from the target representation to the diffusion features (iREPA) yields consistent convergence-speed improvements. See Appendix C for additional results across diverse settings.

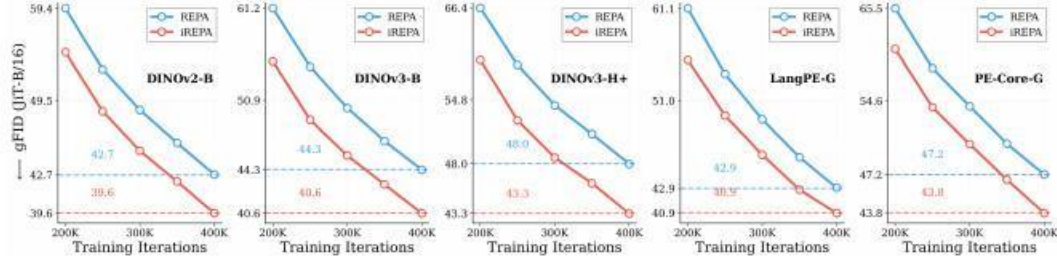


Figure 9: **Convergence comparison for pixel-space diffusion (JiT).** FID convergence curves comparing REPA and iREPA under the pixel-space JiT model (Li & He, 2025b) across vision encoders. Amplifying the transfer of spatial structure consistently accelerates convergence across encoders for pixel-space diffusion. All results are reported using JiT-B/16 (Li & He, 2025b) with batch size 256 and without classifier-free guidance. See Table 10 for further results.

5 RELATED WORK

We summarize the most directly relevant literature here and defer an extended discussion to Appendix H.

Representation alignment for generation. A growing body of work leverages external representations to improve diffusion training (Pernias et al., 2023; Fuest et al., 2024). In particular, Yu et al. (2024); Yao & Wang (2025); Leng et al. (2025a;b); Kouzelis et al. (2025) demonstrate substantial gains from aligning the internal features of a diffusion model with the clean-image features of a pretrained vision encoder. This principle has been extended to video generation (Zhang et al., 2025) and 3D generation (Wu et al., 2025), and Ma et al. (2025a) apply it to the training of unified models. Despite these empirical successes, the precise mechanism by which self-supervised features improve diffusion training remains poorly characterized. This work identifies which property of the target representation is responsible for the improvement and translates that finding into an improved training recipe.

The spatial-global information trade-off in pretrained vision encoders. Several recent studies characterize the trade-off between global and spatial information in pretrained vision encoders (Bolya et al., 2025; Siméoni et al., 2025). Siméoni et al. (2025) show that continued self-supervised training increases the similarity between the global CLS token and the patch tokens, thereby degrading performance on dense spatial tasks, and Bolya et al. (2025); Heinrich et al. (2024) explicitly train spatially tuned encoders for such tasks. We establish that, for generation, the spatial structure of an encoder is more consequential than its global information, and we hope this motivates further work on selecting and training external representations tailored to generation.

Method	DINOv2-B			DINOv3-B			WebSSL-1B			PE-Core-G		
	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓
Baseline REPA	19.06	70.3	5.83	21.47	63.4	6.19	26.10	53.0	6.90	32.35	42.7	6.70
iREPA (w/o spatial norm)	18.52	73.3	6.11	17.76	74.7	5.81	21.17	64.6	6.27	24.97	57.4	6.21
iREPA (w/o conv proj)	17.66	72.8	6.03	18.28	70.8	6.18	18.44	71.0	6.22	21.72	61.5	6.26
iREPA (full)	16.96	77.9	6.26	16.26	78.8	6.14	16.66	77.5	6.18	18.19	75.0	6.03

Table 2: **Ablation on the individual components.** Across all encoders, each component in isolation, the spatial normalization layer and the convolutional projection layer (§4), yields significant convergence-speed gains, and their combination is optimal. All results reported at 100K steps using SiT-XL/2.

Encoder	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑
WebSSL-1B	52.8	26.5	5.20	0.620	0.585
+iREPA-E	87.0	13.2	5.28	0.699	0.598
PE-G	50.9	25.9	5.68	0.612	0.576
+iREPA-E	80.0	16.4	5.40	0.667	0.616
DINOv3-B	82.2	14.4	4.68	0.694	0.596
+iREPA-E	93.6	11.7	4.57	0.703	0.613
DINOv2-B	87.5	12.9	5.40	0.708	0.586
+iREPA-E	91.3	12.1	4.86	0.712	0.602

(a) REPA-E (SiT-XL/2)

Encoder	w/o CFG				w/ CFG			
	4 NFE		1 NFE		4 NFE		1 NFE	
	IS↑	FID↓	IS↑	FID↓	IS↑	FID↓	IS↑	FID↓
WebSSL-1B	27.22	51.40	24.14	58.67	87.85	16.59	69.06	23.74
+iREPA	31.48	45.67	27.33	55.70	100.74	13.89	78.67	20.69
DINOv3-B	28.36	49.58	25.47	57.01	93.30	15.56	72.41	22.55
+iREPA	33.63	44.52	29.67	53.75	124.54	11.05	98.93	17.32
DINOv2-B	28.77	48.87	25.43	56.63	94.40	15.28	71.95	22.17
+iREPA	33.63	44.52	29.67	53.75	111.44	12.59	90.25	18.62

(b) MeanFlow w/ REPA (SiT-B/2)

Table 3: **Variation in training recipes.** Applying our spatial modifications on top of REPA-E (a) and MeanFlow with REPA (b) yields consistent gains. For MeanFlow, results are reported at 1 and 4 NFEs, using a CFG scale of 2.0. All results are obtained at 100K training iterations.

6 CONCLUSION

This work investigates which property of a target representation governs the effectiveness of representation alignment: its global information or its spatial structure. Through a large-scale empirical analysis, we establish that spatial structure, not global information, is the determining factor. We corroborate this conclusion by introducing two minimal modifications that amplify the transfer of spatial information from the target representation to the diffusion features. The resulting method, iREPA, consistently accelerates the convergence of REPA across variation in vision encoders and training recipes. We hope these results motivate a reexamination of the operating mechanism of representation alignment and of how it can be more effectively exploited for the training of generative models.

Vision Encoder	Steps	w/o CFG					w/ CFG				
		IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
DINOv2-B	100K	69.20	19.3	5.89	0.64	0.61	157.2	6.35	5.91	0.769	0.536
+IREPA	100K	77.92	16.9	6.26	0.66	0.61	179.3	5.15	6.23	0.783	0.544
DINOv2-B	400K	127.4	7.76	5.06	0.70	0.66	263.0	1.98	4.60	0.799	0.610
+IREPA	400K	128.6	7.52	4.89	0.71	0.65	268.8	1.93	4.59	0.799	0.600
DINOv3-B	100K	63.64	21.4	6.14	0.63	0.60	144.0	7.57	6.09	0.762	0.526
+IREPA	100K	78.79	16.2	6.14	0.66	0.61	181.9	4.87	6.10	0.780	0.547
DINOv3-B	400K	126.7	8.10	5.06	0.70	0.66	261.2	1.99	4.58	0.799	0.609
+IREPA	400K	132.9	7.13	4.93	0.71	0.66	272.4	1.89	4.58	0.799	0.600
WebSSL-1B	100K	53.87	25.5	6.57	0.61	0.59	124.0	9.59	6.37	0.756	0.515
+IREPA	100K	77.47	16.6	6.18	0.66	0.61	177.6	5.09	6.11	0.787	0.538
WebSSL-1B	400K	116.9	9.39	5.14	0.70	0.64	250.5	2.24	4.61	0.809	0.580
+IREPA	400K	130.8	7.48	4.91	0.70	0.65	271.6	1.90	4.58	0.798	0.609
PE-Core-G	100K	42.74	32.3	6.70	0.57	0.59	97.2	14.1	6.56	0.714	0.525
+IREPA	100K	75.01	18.1	6.03	0.64	0.61	176.8	5.66	6.08	0.771	0.544
PE-Core-G	400K	109.4	10.4	5.00	0.69	0.64	238.4	2.44	4.57	0.805	0.585
+IREPA	400K	132.0	7.78	5.02	0.70	0.65	275.4	1.93	4.59	0.796	0.606

Table 4: **Results across encoders, with and without classifier-free guidance.** SiT-XL/2 is used as the base model. See Appendix E for detailed SiT-B/2 and SiT-L/2 results across all encoders.

REPRODUCIBILITY STATEMENT

The complete set of implementation details and hyperparameters is specified in Appendix G. Our code, model checkpoints, and analysis results will be released upon acceptance.

REFERENCES

- Kristian Abstreiter, Sarthak Mittal, Stefan Bauer, Bernhard Schölkopf, and Arash Mehrjou. Diffusion-based representation learning. In *ICML 2021 Workshop on Unsupervised Reinforcement Learning*, 2021.
- Stability AI. Improved autoencoders ... <https://huggingface.co/stabilityai/sd-vae-ft-mse>, n.d. Accessed: April 11, 2025.
- Anthropic. Claude code. <https://claude.ai/code>, 2025.
- Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 15619–15629, 2023.
- Fan Bao, Shen Nie, Kaiwen Xue, Chongxuan Li, Shi Pu, Yaole Wang, Gang Yue, Yue Cao, Hang Su, and Jun Zhu. All are worth words: A vit backbone for diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 22669–22679, 2023.
- Herbert Bay, Tinne Tuytelaars, and Luc Van Gool. Surf: Speeded up robust features. In *Computer Vision—ECCV 2006: 9th European Conference on Computer Vision, Graz, Austria, May 7–13, 2006. Proceedings, Part I 9*, pp. 404–417. Springer, 2006.
- Daniel Bolya, Po-Yao Huang, Peize Sun, Jang Hyun Cho, Andrea Madotto, Chen Wei, Tengyu Ma, Jiale Zhi, Jathushan Rajasegaran, Hanoona Rasheed, Junke Wang, Marco Monteiro, Hu Xu, Shiyu Dong, Nikhila Ravi, Daniel Li, Piotr Dollár, and Christoph Feichtenhofer. Perception encoder: The best visual embeddings are not at the output of the network. *arXiv:2504.13181*, 2025.
- Shoufa Chen, Mengmeng Xu, Jiawei Ren, Yuren Cong, Sen He, Yanping Xie, Nicu Sebe, and Wai Lam. Gentron: Diffusion transformers for image and video generation. *arXiv preprint arXiv:2312.04557*, 2023.
- Shoufa Chen, Chongjian Ge, Shilong Zhang, Peize Sun, and Ping Luo. Pixelflow: Pixel-space generative models with flow. *arXiv preprint arXiv:2504.07963*, 2025.

- Xinlei Chen, Saining Xie, and Kaiming He. An empirical study of training self-supervised vision transformers. In *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 9640–9649, 2021.
- Xinlei Chen, Zhuang Liu, Saining Xie, and Kaiming He. Deconstructing denoising diffusion models for self-supervised learning. *arXiv preprint arXiv:2401.14404*, 2024.
- Jang Hyun Cho, Andrea Madotto, Effrosyni Mavroudi, Triantafyllos Afouras, Tushar Nagarajan, Muhammad Maaz, Yale Song, Tengyu Ma, Shuming Hu, Hanoona Rasheed, Peize Sun, Po-Yao Huang, Daniel Bolya, Suyog Jain, Miguel Martin, Huiyu Wang, Nikhila Ravi, Shashank Jain, Temmy Stark, Shane Moon, Babak Damavandi, Vivian Lee, Andrew Westbury, Salman Khan, Philipp Krähenbühl, Piotr Dollár, Lorenzo Torresani, Kristen Grauman, and Christoph Feichtenhofer. Perceptionlm: Open-access data and models for detailed visual understanding. *arXiv:2504.13180*, 2025.
- N. Dalal and B. Triggs. Histograms of oriented gradients for human detection. In *2005 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR’05)*, volume 1, pp. 886–893 vol. 1, 2005. doi: 10.1109/CVPR.2005.177.
- Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *2009 IEEE conference on computer vision and pattern recognition*, pp. 248–255. Ieee, 2009.
- Prafulla Dhariwal and Alexander Nichol. Diffusion models beat gans on image synthesis. *Advances in neural information processing systems*, 34:8780–8794, 2021.
- Patrick Esser, Sumith Kulal, Andreas Blattmann, Rahim Entezari, Jonas Müller, Harry Saini, Yam Levi, Dominik Lorenz, Axel Sauer, Frederic Boesel, et al. Scaling rectified flow transformers for high-resolution image synthesis. In *Forty-first International Conference on Machine Learning*, 2024.
- David Fan, Shengbang Tong, Jiachen Zhu, Koustuv Sinha, Zhuang Liu, Xinlei Chen, Michael Rabbat, Nicolas Ballas, Yann LeCun, Amir Bar, et al. Scaling language-free visual representation learning. *arXiv preprint arXiv:2504.01017*, 2025.
- Michael Fuest, Pingchuan Ma, Ming Gui, Johannes Schusterbauer, Vincent Tao Hu, and Bjorn Ommer. Diffusion models and representation learning: A survey. *arXiv preprint arXiv:2407.00783*, 2024.
- Zhengyang Geng, Mingyang Deng, Xingjian Bai, J Zico Kolter, and Kaiming He. Mean flows for one-step generative modeling. *arXiv preprint arXiv:2505.13447*, 2025.
- Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. Generative adversarial networks. *Communications of the ACM*, 63(11):139–144, 2020.
- Greg Heinrich, Mike Ranzinger, Hongxu Yin, Yao Lu, Jan Kautz, Andrew Tao, Bryan Catanzaro, and Pavlo Molchanov. Radiov2.5: Improved baselines for agglomerative vision foundation models, 2024. URL <https://arxiv.org/abs/2412.07679>.
- Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. Gans trained by a two time-scale update rule converge to a local nash equilibrium. *Advances in neural information processing systems*, 30, 2017.
- Jing Huang, S.R. Kumar, M. Mitra, Wei-Jing Zhu, and R. Zabih. Image indexing using color correlograms. In *Proceedings of IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, pp. 762–768, 1997. doi: 10.1109/CVPR.1997.609412.
- Drew A Hudson, Daniel Zoran, Mateusz Malinowski, Andrew K Lampinen, Andrew Jaegle, James L McClelland, Loic Matthey, Felix Hill, and Alexander Lerchner. Soda: Bottleneck diffusion models for representation learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 19461–19471, 2024.

- Sadeep Jayasumana, Srikumar Ramalingam, Andreas Veit, Daniel Glasner, Ayan Chakrabarti, and Sanjiv Kumar. Rethinking fid: Towards a better evaluation metric for image generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 9307–9315, June 2024.
- Dengyang Jiang, Mengmeng Wang, Liuzhuozheng Li, Lei Zhang, Haoyu Wang, Wei Wei, Guang Dai, Yanning Zhang, and Jingdong Wang. No other representation component is needed: Diffusion transformers can provide representation guidance by themselves. *arXiv preprint arXiv:2505.02831*, 2025.
- Jiatao Jing, Gu Wu, Yuanpeng Zhou, Yuxin Xu, Zixun Sun, Xinhao Ni, Yazhou Wang, Chunmeng Li, Hao Tang, Ruihua Chen, et al. Dart: Denoising autoregressive transformers for scalable text-to-image generation. *arXiv preprint arXiv:2410.08159*, 2024.
- Minguk Kang, Richard Zhang, Connelly Barnes Nga Tran, Saurabh Kar, and Jun-Yan Zhu. Distilling diffusion models into conditional gans. *arXiv preprint arXiv:2405.05967*, 2024.
- Diederik P Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2014.
- Theodoros Kouzelis, Efstathios Karypidis, Ioannis Kakogeorgiou, Spyros Gidaris, and Nikos Komodakis. Boosting generative image modeling via joint image-feature synthesis. *arXiv preprint arXiv:2504.16064*, 2025.
- Tuomas Kynkäänniemi, Tero Karras, Samuli Laine, Jaakko Lehtinen, and Timo Aila. Improved precision and recall metric for assessing generative models. *Advances in neural information processing systems*, 32, 2019.
- Tuomas Kynkäänniemi, Tero Karras, Miika Aittala, Timo Aila, and Jaakko Lehtinen. The role of imagenet classes in fréchet inception distance. In *International Conference on Learning Representations*, 2023.
- Xingjian Leng, Jaskirat Singh, Yunzhong Hou, Zhenchang Xing, Saining Xie, and Liang Zheng. Repa-e: Unlocking vae for end-to-end tuning with latent diffusion transformers. *arXiv preprint arXiv:2504.10483*, 2025a.
- Xingjian Leng, Jaskirat Singh, Ryan Murdock, Ethan Smith, Rebecca Li, Yunzhong Hou, Zhenchang Xing, Saining Xie, and Liang Zheng. Family of End-to-End Tuned VAEs for Supercharging T2I Diffusion Transformers. <https://end2end-diffusion.github.io/repa-e-t2i/>, 2025b.
- Tianhong Li and Kaiming He. Back to basics: Let denoising generative models denoise. *arXiv preprint arXiv:2511.13720*, 2025a.
- Tianhong Li and Kaiming He. Back to basics: Let denoising generative models denoise. *arXiv preprint arXiv:2511.13720*, 2025b.
- I Loshchilov. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101*, 2017.
- David G Lowe. Object recognition from local scale-invariant features. In *Proceedings of the seventh IEEE international conference on computer vision*, volume 2, pp. 1150–1157. Ieee, 1999.
- Nanye Ma, Mark Goldstein, Michael S Albergo, Nicholas M Boffi, Eric Vanden-Eijnden, and Saining Xie. Sit: Exploring flow and diffusion-based generative models with scalable interpolant transformers. In *European Conference on Computer Vision*, pp. 23–40. Springer, 2024.
- Yiyang Ma, Xingchao Liu, Xiaokang Chen, Wen Liu, Chengyue Wu, Zhiyu Wu, Zizheng Pan, Zhenda Xie, Haowei Zhang, Xingkai Yu, et al. Janusflow: Harmonizing autoregression and rectified flow for unified multimodal understanding and generation. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 7739–7751, 2025a.
- Zehong Ma, Longhui Wei, Shuai Wang, Shiliang Zhang, and Qi Tian. Deco: Frequency-decoupled pixel diffusion for end-to-end image generation. *arXiv preprint arXiv:2511.19365*, 2025b.

- Charlie Nash, Jacob Menick, Sander Dieleman, and Peter Battaglia. Generating images with sparse representations. In *International Conference on Machine Learning*, pp. 7958–7968. PMLR, 2021.
- OpenAI. Gpt-5. <https://openai.com>, 2025.
- Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. Dinov2: Learning robust visual features without supervision. *Transactions on Machine Learning Research Journal*, pp. 1–31, 2024.
- Sayak Paul. Cmmd-pytorch: Pytorch implementation of cmmd metric. <https://github.com/sayakpaul/cmmd-pytorch>, 2024. Accessed: 2025.
- William Peebles and Saining Xie. Scalable diffusion models with transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 4195–4205, 2023.
- Pablo Pernias, Dominic Rampas, Mats Leon Richter, Christopher Pal, and Marc Aubreville. Würstchen: An efficient architecture for large-scale text-to-image diffusion models. In *The Twelfth International Conference on Learning Representations*, 2023.
- Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pp. 8748–8763. PMLR, 2021.
- Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, Eric Mintun, Junting Pan, Kalyan Vasudev Alwala, Nicolas Carion, Chao-Yuan Wu, Ross Girshick, Piotr Dollár, and Christoph Feichtenhofer. Sam 2: Segment anything in images and videos. *arXiv preprint arXiv:2408.00714*, 2024. URL <https://arxiv.org/abs/2408.00714>.
- Ethan Rublee, Vincent Rabaud, Kurt Konolige, and Gary Bradski. Orb: An efficient alternative to sift or surf. In *2011 International conference on computer vision*, pp. 2564–2571. Ieee, 2011.
- Tim Salimans, Ian Goodfellow, Wojciech Zaremba, Vicki Cheung, Alec Radford, and Xi Chen. Improved techniques for training gans. *Advances in neural information processing systems*, 29, 2016.
- Axel Sauer, Katja Schwarz, and Andreas Geiger. Projected gans converge faster. In *Advances in Neural Information Processing Systems*, volume 34, pp. 17480–17492, 2021.
- Axel Sauer, Katja Schwarz, and Andreas Geiger. Stylegan-xl: Scaling stylegan to large diverse datasets. In *ACM SIGGRAPH 2022 conference proceedings*, pp. 1–10, 2022.
- Axel Sauer, Tero Karras, Samuli Laine, Andreas Geiger, and Timo Aila. Stylegan-t: Unlocking the power of gans for fast large-scale text-to-image synthesis. *arXiv preprint arXiv:2301.09515*, 2023.
- Axel Sauer, Dominik Lorenz, Andreas Blattmann, and Robin Rombach. Adversarial diffusion distillation. In *European Conference on Computer Vision*, pp. 218–234. Springer, 2024.
- Eli Shechtman and Michal Irani. Matching local self-similarities across images and videos. In *2007 IEEE Conference on Computer Vision and Pattern Recognition*, pp. 1–8, 2007. doi: 10.1109/CVPR.2007.383198.
- Oriane Siméoni, Huy V. Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, Cijo Jose, Vasil Khalidov, Marc Szafraniec, Seungeun Yi, Michaël Ramamonjisoa, Francisco Massa, Daniel Haziza, Luca Wehrstedt, Jianyuan Wang, Timothée Darcet, Théo Moutakanni, Leonel Sentana, Claire Roberts, Andrea Vedaldi, Jamie Tolan, John Brandt, Camille Couprie, Julien Mairal, Hervé Jégou, Patrick Labatut, and Piotr Bojanowski. DINOv3, 2025. URL <https://arxiv.org/abs/2508.10104>.
- Karen Simonyan and Andrew Zisserman. Very deep convolutional networks for large-scale image recognition. *arXiv preprint arXiv:1409.1556*, 2014.

- George Stein, Jesse C. Cresswell, Rasa Hosseinzadeh, Yi Sui, Brendan Leigh Ross, Valentin Villecroze, Zhaoyan Liu, Anthony L. Caterini, J. Eric T. Taylor, and Gabriel Loaiza-Ganem. Exposing flaws of generative model evaluation metrics and their unfair treatment of diffusion models. In *Advances in Neural Information Processing Systems*, 2023.
- Shitao Tang, Fuyang Chen, Aviral Joshi Bose, Kun Zhang, Shengjie Bi, Zhenyu Wang, Jiacheng Feng, Haozhe Cai, Alexander Liu, Antonio Torralba, et al. Ardit: Autoregressive diffusion transformers. *arXiv preprint arXiv:2406.00843*, 2024.
- Dmitry Ulyanov, Andrea Vedaldi, and Victor Lempitsky. Instance normalization: The missing ingredient for fast stylization. *arXiv preprint arXiv:1607.08022*, 2016.
- Runqian Wang and Kaiming He. Diffuse and disperse: Image generation with representation regularization. *arXiv preprint arXiv:2506.09027*, 2025.
- Shuai Wang, Ziteng Gao, Chenhui Zhu, Weilin Huang, and Limin Wang. Pixnerd: Pixel neural field diffusion. *arXiv preprint arXiv:2507.23268*, 2025a.
- Shuai Wang, Zhi Tian, Weilin Huang, and Limin Wang. Ddt: Decoupled diffusion transformer. *arXiv preprint arXiv:2504.05741*, 2025b.
- Ziqiao Wang, Wangbo Zhao, Yuhao Zhou, Zekai Li, Zhiyuan Liang, Mingjia Shi, Xuanlei Zhao, Pengfei Zhou, Kaipeng Zhang, Zhangyang Wang, Kai Wang, and Yang You. Repa works until it doesn't: Early-stopped, holistic alignment supercharges diffusion training. *arXiv preprint arXiv:2505.16792*, 2025c.
- Haoyu Wu, Diankun Wu, Tianyu He, Junliang Guo, Yang Ye, Yueqi Duan, and Jiang Bian. Geometry forcing: Marrying video diffusion and 3d representation for consistent world modeling. *arXiv preprint arXiv:2507.07982*, 2025.
- Mengping Yang, Ceyuan Yang, Yichi Zhang, Qingyan Bai, Yujun Shen, and Bo Dai. Revisiting the evaluation of image synthesis with gans. In *Advances in Neural Information Processing Systems*, 2023.
- Jingfeng Yao and Xinggang Wang. Reconstruction vs. generation: Taming optimization dilemma in latent diffusion models. *arXiv preprint arXiv:2501.01423*, 2025.
- Sihyun Yu, Sangkyung Kwak, Huiwon Jang, Jongheon Jeong, Jonathan Huang, Jinwoo Shin, and Saining Xie. Representation alignment for generation: Training diffusion transformers is easier than you think. *arXiv preprint arXiv:2410.06940*, 2024.
- Xiangdong Zhang, Jiaqi Liao, Shaofeng Zhang, Fanqing Meng, Xiangpeng Wan, Junchi Yan, and Yu Cheng. Videorepa: Learning physics for video generation through relational alignment with foundation models. *arXiv preprint arXiv:2505.23656*, 2025.

A DETAILED CORRELATION ANALYSIS

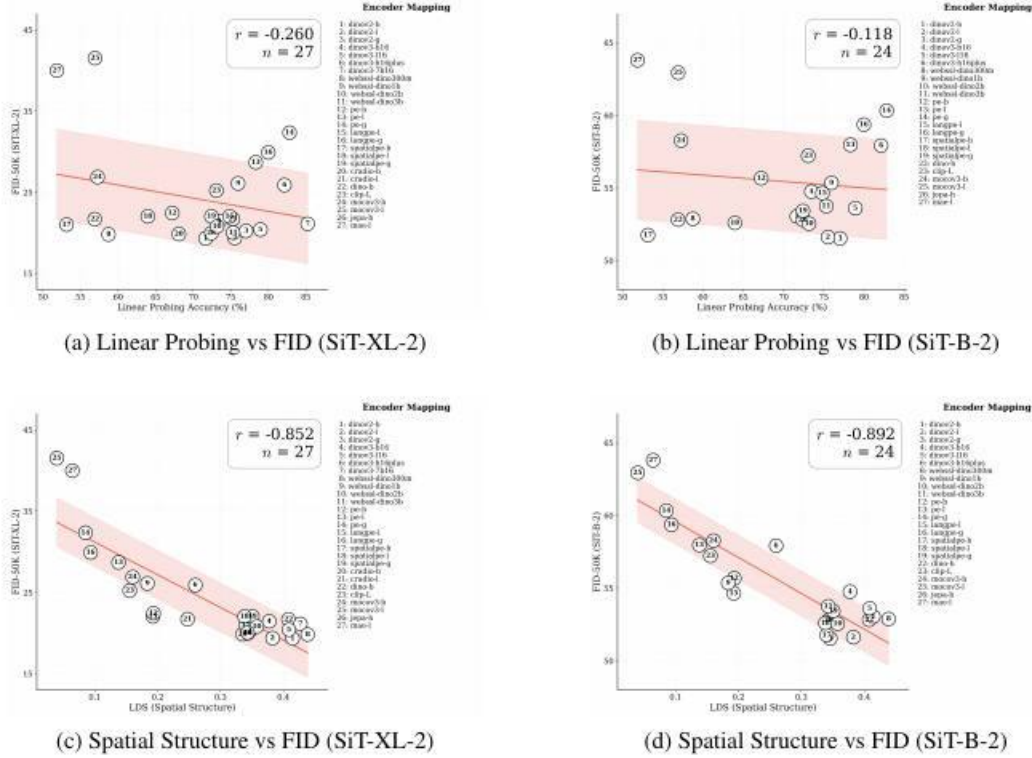


Figure 10: **Detailed correlation analysis with per-encoder labels.** The top row plots linear-probing accuracy against FID for the SiT-XL-2 and SiT-B-2 models; the bottom row plots spatial structure (LDS) against FID. Each point is annotated with its encoder index (see legend). At both model scales, the spatial structure metric exhibits a substantially stronger correlation with generation quality than linear-probing accuracy.

B SPATIAL SELF-SIMILARITY METRICS



Figure 11: **Illustration of the semantic-region self-similarity (SRSS) metric.** For each image, we compute a mask with SAM2 and sample anchor–positive–negative triplets. The SSM is the difference between the cosine similarity of anchor–positive pairs (inside the mask) and that of anchor–negative pairs (outside the mask). A larger value indicates that patches within a common semantic region are more similar than patches across unrelated regions, i.e., stronger preservation of spatial structure.

Setup. Given an image $\mathcal{I}$ and a vision encoder $\mathcal{E}$, we extract patch tokens $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ over $T = H \times W$ patches, with indices $t \in [T]$ arranged on an $H \times W$ lattice. Let $d : [T] \times [T] \rightarrow \mathbb{N}$ denote the Manhattan distance on this lattice, and let the (cosine) self-similarity kernel be

$$K_{\mathcal{X}}(t, t') = \frac{\langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle}{\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2} \in [-1, 1],$$

a standard local self-similarity measure (Shechtman & Irani, 2007).

- *Patch token representation:* $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ where $T = H \times W$ is the spatial grid of patches (patch index set $[T]$). Also let $d : [T] \times [T] \rightarrow \mathbb{N}$ be Manhattan distance on the $H \times W$ grid.
- *Self-similarity kernel:* $K : [T] \times [T] \rightarrow \mathbb{R}$ measuring self-similarity (Shechtman & Irani, 2007) between patch tokens. We use the cosine kernel $K_{\mathcal{X}}(t, t') = \langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle / (\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2)$.
- *Spatial self-similarity metric:* a functional $m : \mathcal{K} \times \mathcal{D} \rightarrow \mathbb{R}$ summarizing how the self-similarity K between patch tokens varies with lattice distance d . Larger values indicate stronger spatial organization, i.e., nearby patches are more mutually similar than distant patches.

We now define several tractable metrics for quantifying the spatial self-similarity structure of a target representation.

Local vs. Distant Similarity (LDS). As a first instantiation, we adopt a correlogram-contrast (local versus distant) metric (Huang et al., 1997):

$$\text{LDS}(\mathcal{X}) = \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') < r_{\text{near}}] - \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') \geq r_{\text{far}}],$$

where r_{near} and r_{far} are hyperparameters, set by default to $r_{\text{near}} = r_{\text{far}} = H/2$; the resulting correlations are empirically insensitive to these values. A larger LDS indicates that, *on average*, spatially proximate patches are more similar than distant ones.

Correlation Decay Slope (CDS). Given the patch tokens $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$, we compute the spatial correlogram $g_{\mathcal{X}}(\delta) = \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') = \delta]$ over distances $\delta \in \{1, \dots, \Delta\}$, fit a least-squares line $\hat{g}_{\mathcal{X}}(\delta) \approx \alpha + \beta\delta$, and define:

$$\text{CDS}(\mathcal{X}) = -\hat{\beta}$$

with $\hat{\beta}$ the fitted slope. A larger CDS corresponds to faster similarity decay with distance, indicating stronger spatial organization.

Semantic-Region Self-Similarity (SRSS). For each image, we sample a binary mask $\mathbf{M} \in \mathbb{R}^{H \times W}$ delineating a random object using SAM2 (Ravi et al., 2024), downsampled to $H \times W$. From this mask we sample three points, an anchor, a positive, and a negative. The anchor and positive are drawn from within the mask (and thus share semantics), while the negative is drawn from outside the mask (and is less related to the anchor). If the encoder representation preserves spatial structure, the anchor–positive pair should exhibit higher cosine similarity than the anchor–negative pair. We accordingly define the Spatial Structure Metric (SSM) as:

$$\text{SSM}(\mathbf{P}) = \mathbb{E}_{\text{anchor} \in \mathbf{M}} [\cos(\mathbf{p}_{\text{anchor}}, \mathbf{p}_{\text{pos}}) - \cos(\mathbf{p}_{\text{anchor}}, \mathbf{p}_{\text{neg}})]$$

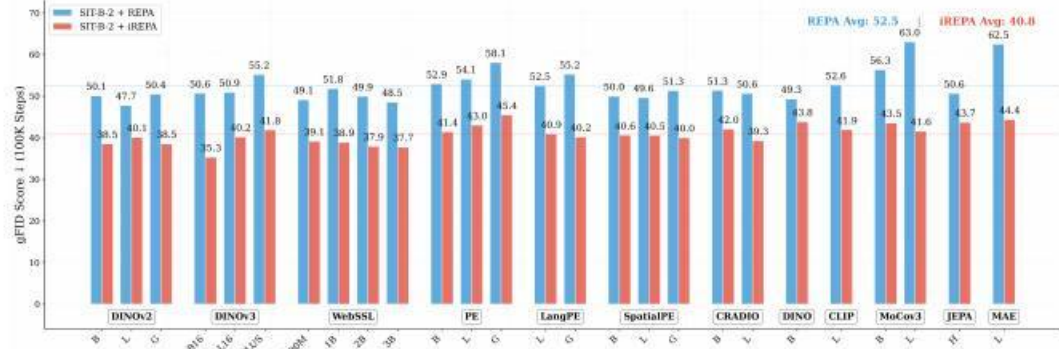
where positive points lie within Manhattan distance $d \leq r_{near}$ of the anchor inside the mask, and negative points lie at distance $d \geq r_{far}$ outside the mask.

RMS Spatial Contrast (RMSC). Finally, we define a contrast metric that measures the dispersion of patch-token representations about their spatial mean. Using ℓ_2 -normalized features $\hat{\mathbf{x}}_t = \mathbf{x}_t / \|\mathbf{x}_t\|_2$ for each patch t , we compute:

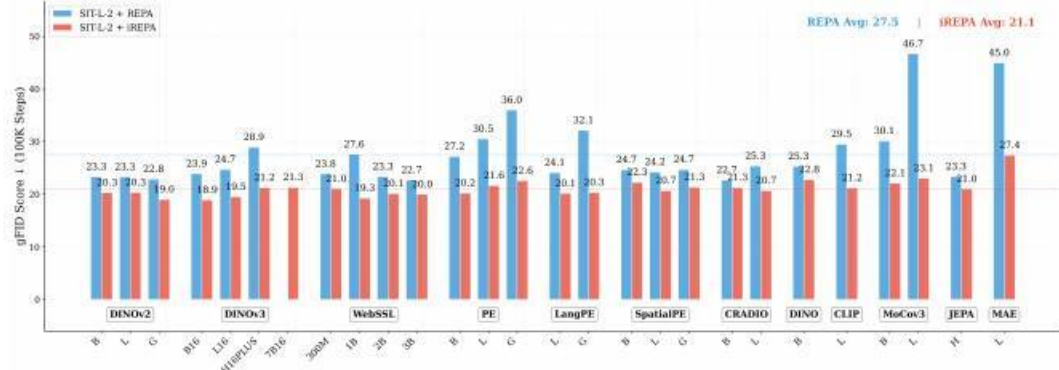
$$\text{RMSC}(\mathcal{X}) = \sqrt{\frac{1}{T} \sum_{t=1}^T \|\hat{\mathbf{x}}_t - \bar{\mathbf{x}}\|_2^2}$$

where $\bar{\mathbf{x}} = \frac{1}{T} \sum_{t=1}^T \hat{\mathbf{x}}_t$ is the mean of the normalized features across all spatial locations. A higher RMSC reflects greater spatial diversity among the features, consistent with preserved spatial structure, whereas a lower RMSC reflects a more uniform feature distribution, indicating a loss of spatial structure (as typically occurs when global information dominates (Siméoni et al., 2025)).

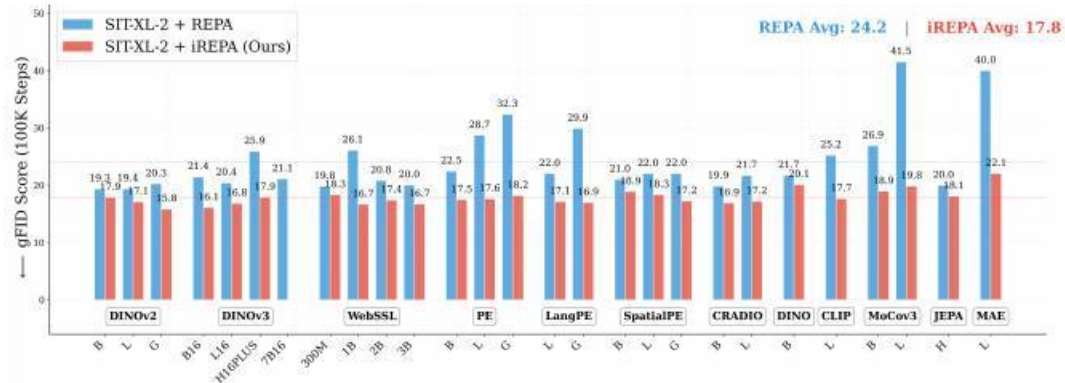
C VISION ENCODER VARIATION



(a) SiT-B-2 (130M parameters)



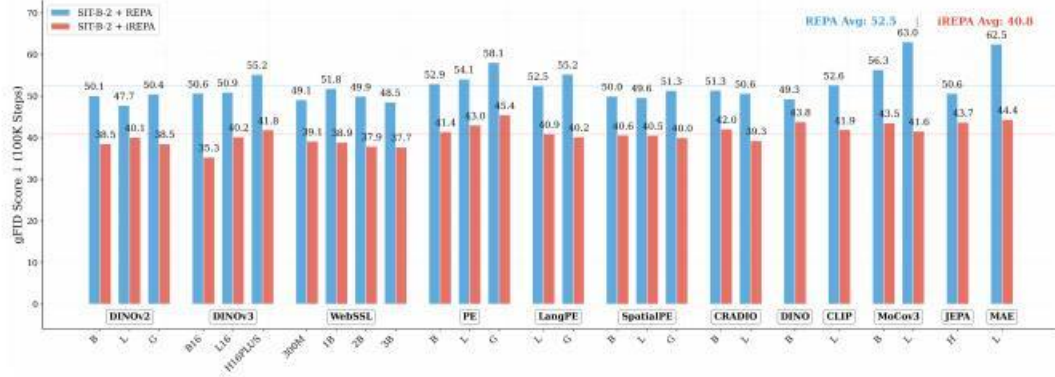
(b) SiT-L-2 (458M parameters)



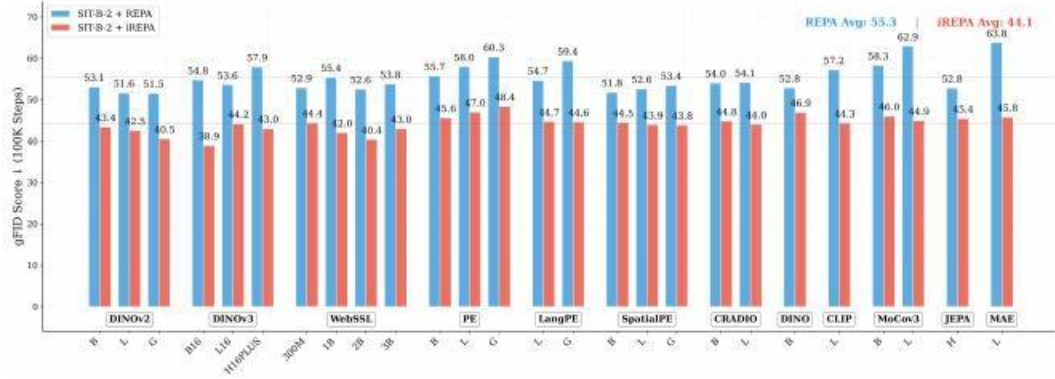
(c) SiT-XL-2 (675M parameters)

Figure 12: **Variation in vision encoders.** Ablation quantifying the iREPA improvement across vision encoders for the SiT-B-2, SiT-L-2, and SiT-XL-2 models. iREPA improves generation quality for every encoder and model size.

D ENCODER DEPTH VARIATION



(a) SiT-B-2 with 6 encoder layers



(b) SiT-B-2 with 8 encoder layers

Figure 13: **Effect of encoder depth on iREPA performance.** SiT-B-2 performance under two encoder depths (6 versus 8 layers) across vision encoders. iREPA improves performance at both depths, with the 8-layer configuration attaining slightly stronger results. Both configurations exhibit substantial FID reductions when the spatial modifications are applied.

E COMPREHENSIVE RESULTS ACROSS VISION ENCODERS

E.1 SiT-B/2 RESULTS

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
CLIP-ViT-L	59.44	24.84	6.46	0.584	0.645
+iREPA	69.70	21.26	6.64	0.601	0.644
DINOv2-B	64.92	22.75	6.54	0.593	0.653
+iREPA	70.88	21.40	6.77	0.597	0.653
DINOv2-L	66.65	22.46	6.53	0.594	0.650
+iREPA	72.41	20.99	6.84	0.600	0.650
DINOv3-B16	62.57	23.91	6.77	0.584	0.647
+iREPA	75.48	19.65	6.68	0.606	0.649
DINOv3-H16+	61.91	24.82	6.57	0.579	0.645
+iREPA	71.76	21.80	6.84	0.592	0.655
DINO-B	58.37	24.71	6.32	0.586	0.635
+iREPA	57.55	24.41	6.43	0.587	0.638
LangPE-G	58.42	25.37	6.32	0.583	0.641
+iREPA	66.88	21.62	6.65	0.604	0.649
MoCov3-B	57.61	25.60	6.25	0.579	0.640
+iREPA	61.41	22.80	6.46	0.596	0.639
PE-B	58.86	25.78	6.44	0.58	0.64
+iREPA	69.50	21.15	6.66	0.601	0.659
PE-G	56.09	26.84	6.45	0.568	0.641
+iREPA	70.43	21.52	6.58	0.598	0.651
SpatialPE-B	59.19	24.28	6.37	0.586	0.646
+iREPA	61.50	22.71	6.41	0.604	0.644
SpatialPE-L	62.13	23.87	6.51	0.59	0.65
+iREPA	68.83	21.07	6.59	0.607	0.651
WebSSL-1B	63.20	23.48	6.33	0.593	0.646
+iREPA	73.54	19.74	6.63	0.612	0.646
WebSSL-2B	66.56	22.27	6.69	0.596	0.654
+iREPA	72.13	20.47	6.89	0.609	0.648
WebSSL-300M	62.06	23.64	6.48	0.590	0.646
+iREPA	66.83	21.46	6.71	0.606	0.653
WebSSL-3B	65.91	22.55	6.53	0.598	0.654
+iREPA	71.98	20.64	6.71	0.607	0.650

Table 5: **SiT-B/2** performance across vision encoders at **400K** iterations. iREPA improves generation quality for every encoder. All baselines are trained with vanilla REPA (Yu et al., 2024).

E.2 SiT-L/2 RESULTS

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
CLIP-ViT-L	107.7	10.6	5.20	0.682	0.643
+iREPA	115.8	9.41	5.15	0.689	0.653
DINOv2-B	113.3	9.61	5.11	0.684	0.653
+iREPA	113.9	9.36	5.11	0.690	0.655
DINOv3-B16	115.3	9.41	5.24	0.687	0.651
+iREPA	118.1	9.06	5.25	0.685	0.650
DINOv3-H16+	110.7	10.6	5.32	0.677	0.652
+iREPA	120.5	9.64	5.29	0.681	0.663
LangPE-G	102.8	11.2	5.13	0.684	0.637
+iREPA	112.7	9.37	5.02	0.690	0.653
MoCov3-B	102.1	11.0	5.14	0.685	0.635
+iREPA	104.7	10.1	5.00	0.690	0.641
PE-G	98.38	12.6	5.31	0.670	0.646
+iREPA	117.1	9.65	5.33	0.683	0.651
WebSSL-1B	107.9	10.5	5.13	0.686	0.644
+iREPA	121.5	8.62	5.01	0.696	0.653

Table 6: **SiT-L/2 performance across vision encoders at 400K iterations.** iREPA improves generation quality for every encoder, with especially strong gains in FID and IS. All baselines are trained with vanilla REPA (Yu et al., 2024).

F ADDITIONAL EXPERIMENTAL RESULTS

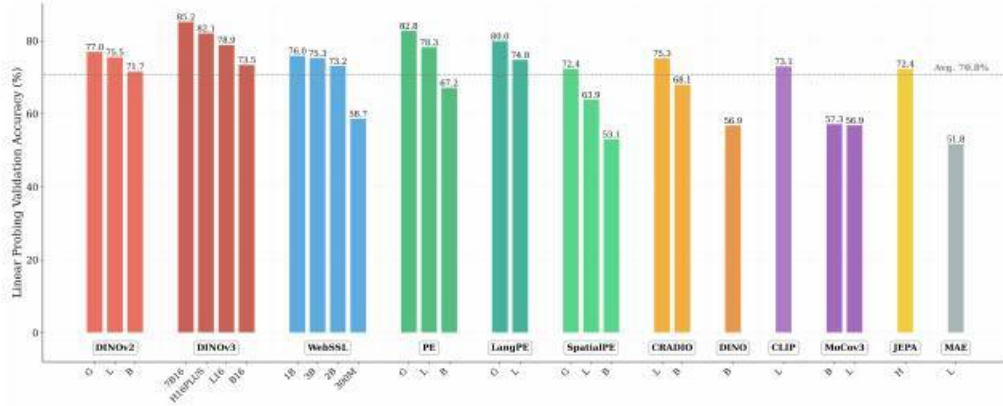


Figure 14: **Global information in the mean of the patch tokens.** In addition to the [CLS] token, the mean of the patch tokens carries substantial global semantic information. Although this raises global performance, it reduces the contrast among individual patch tokens and thereby impedes spatial structure transfer. Removing part of this global information (the mean of the patch tokens) improves spatial structure transfer during representation alignment (§4).

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
DINOv2-B	262.97	1.98	4.60	0.799	0.610
+iREPA (Ours)	268.79	1.93	4.59	0.799	0.600
DINOv3-B	261.18	1.99	4.58	0.799	0.609
+iREPA (Ours)	272.41	1.89	4.58	0.799	0.600
WebSSL-1B	250.53	2.24	4.61	0.809	0.580
+iREPA (Ours)	271.59	1.90	4.58	0.798	0.609
PE-G	238.37	2.44	4.57	0.805	0.585
+iREPA (Ours)	275.36	1.93	4.59	0.796	0.606

Table 7: **Generation quality under classifier-free guidance.** Comparison of REPA and iREPA under CFG (scale 2.0) across vision encoders. All experiments use SiT-XL/2 trained for 400K iterations with 250 sampling steps. iREPA improves both IS and FID for every encoder.

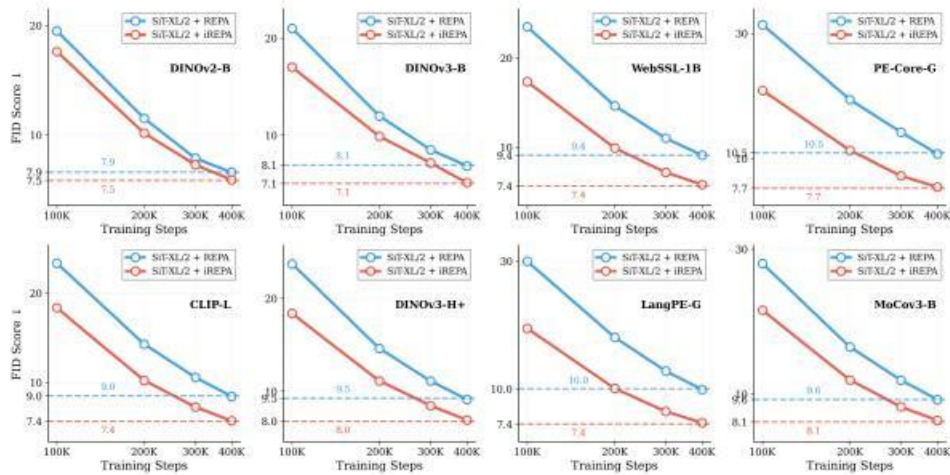


Figure 15: **Amplifying spatial features consistently accelerates convergence.** Results for SiT-XL/2 under REPA and iREPA.

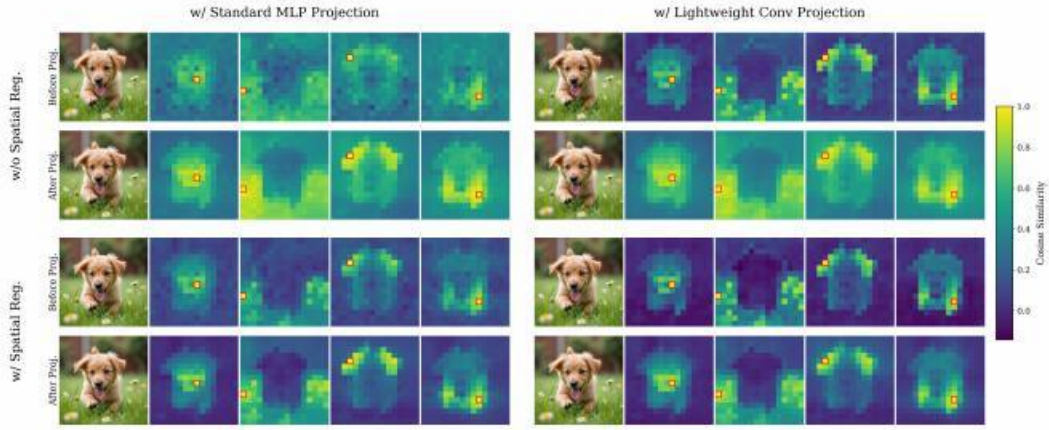


Figure 16: **Visualizing the effect of the two minimal modifications that enhance spatial feature transfer.** **First**, the standard MLP projection layer (top-left) discards spatial information when mapping the pretrained encoder features (after projection) onto the diffusion model (before projection); replacing it with a single convolutional layer preserves this information (top-right). **Second**, vision-encoder features frequently exhibit limited spatial contrast (§2), so that tokens in one semantic region (*e.g.*, dog) attain appreciable cosine similarity with unrelated tokens (*e.g.*, background). We address this with a spatial normalization layer that increases the spatial contrast of the learned representation, thereby improving generation. The best results are obtained when both modifications are applied (see Table 2).

G IMPLEMENTATION DETAILS

Training setup. Unless otherwise specified, we follow the training protocol of REPA (Yu et al., 2024). All models are trained on the ImageNet (Deng et al., 2009) training split. For pre-processing we adopt the ADM (Dhariwal & Nichol, 2021) protocol, center-cropping and resizing each image to 256×256 resolution. Throughout diffusion training and inference we use the stabilityai/sd-vae-ft-mse VAE (AI, n.d.). The spatial normalization layer uses $\gamma \in [0.6, 0.8]$, and the projection layer is a convolution with kernel size 3 and padding 1. Optimization uses AdamW (Kingma & Ba, 2014; Loshchilov, 2017) with a constant learning rate of 1×10^{-4} and a global batch size of 256. Training employs bfloat16 mixed precision and torch.compile for acceleration, together with gradient clipping and an exponential moving average (EMA) of the generative model for optimization stability.

For the REPA-E (Leng et al., 2025a) and JiT (Li & He, 2025a) experiments we use the official open-source implementations. For the MeanFlow (Geng et al., 2025) experiments we obtained the implementation (including REPA) from the original authors and adapted it to incorporate the two iREPA modifications (§4).

Evaluation. For image-generation evaluation we adhere strictly to the ADM protocol (Dhariwal & Nichol, 2021). We report generation quality via the Fréchet inception distance (gFID) (Heusel et al., 2017), structural FID (sFID) (Nash et al., 2021), inception score (IS) (Salimans et al., 2016), and precision (Prec.) and recall (Rec.) (Kynkäänniemi et al., 2019), each computed over 50K generated images. Sampling follows REPA (Yu et al., 2024), using the SDE Euler–Maruyama sampler with 250 steps; for JiT (Li & He, 2025a) we use 50 inference steps, matching the official implementation.

H MORE DISCUSSION ON RELATED WORK

Classical spatial features in computer vision. Spatial feature extraction has long been foundational to computer vision. Classical descriptors such as SIFT (Lowe, 1999), HOG (Dalal & Triggs, 2005), SURF (Bay et al., 2006), and ORB (Rublee et al., 2011) furnish robust local descriptors for tasks ranging from object detection to image matching. Although these handcrafted descriptors capture local spatial patterns and geometric invariances effectively, recent generative modeling has concentrated on representations from modern self-supervised methods with strong global classification performance, such as DINOv2 (Oquab et al., 2024) and CLIP (Radford et al., 2021). Our findings support a different perspective: because the preservation of spatial structure is decisive for generation quality, even classical spatial descriptors can improve diffusion training when properly aligned. This indicates the potential of the full spectrum of spatial feature extractors, from handcrafted descriptors to learned representations, provided they maintain strong spatial coherence.

Pretrained visual encoders for generative models. Pretrained visual encoders have served generative models in several roles: as discriminators that accelerate GAN convergence (Goodfellow et al., 2020; Sauer et al., 2021; 2022; 2023; Radford et al., 2021), as teachers in adversarial distillation for diffusion models (Sauer et al., 2024; Kang et al., 2024), and, more recently, as alignment targets. In GANs (Goodfellow et al., 2020), pretrained features have both improved convergence speed and enabled scaling to large datasets, as demonstrated by StyleGAN-XL (Sauer et al., 2022) and StyleGAN-T (Sauer et al., 2023) with CLIP (Radford et al., 2021) features. For diffusion models, adversarial distillation exploits pretrained encoders to steer student networks toward higher-fidelity samples, with clear gains in perceptual quality. In particular, REPA (Yu et al., 2024) aligns diffusion features with external encoders, establishing that representation alignment improves both generation convergence and quality. Building on this line of work, we focus not on the discriminative capability of the encoder but specifically on its preservation of spatial structure.

Denosing transformers. Transformer architectures have become the dominant backbone for scalable generative modeling, in formulations that include diffusion transformers (Peebles & Xie, 2023) and flow-matching variants (Ma et al., 2024). Recent architectures such as GenTron (Chen et al., 2023) scale transformers beyond 3B parameters for text-to-image synthesis, U-ViT (Bao et al., 2023) shows that pure transformer backbones without U-Net inductive biases attain competitive performance, and ARDiT (Tang et al., 2024) and DART (Jing et al., 2024) explore autoregressive formulations that combine denoising with sequential generation to trade off quality against speed. Despite these advances, training such models from scratch remains computationally expensive, frequently requiring

millions of iterations to reach adequate generation quality. Although representation-alignment methods such as REPA (Yu et al., 2024) demonstrate that pretrained features can substantially accelerate convergence, the mechanism underlying this acceleration has been unclear. Our analysis attributes the benefit to the preservation of spatial structure rather than to semantic alignment, which explains why certain encoders accelerate training more than others and informs the design of more effective alignment strategies. While (Wang et al., 2025c) propose improvements to REPA training through early stopping, they hypothesize that “REPA predominantly distills global semantic information while leaving structural information untapped.” In contrast, our analysis reveals that spatial structure (not global semantic information) already plays a very significant role in the effectiveness of REPA (§2, §3). Thus, while we indeed find that spatial structure remains underexploited (§4) in REPA, we surprisingly find that majority of improvements in REPA are already coming from introducing an inductive bias for spatial structure (not global information).

Denosing as a self-supervised learning task. The relationship between denoising and representation learning has been examined from several angles. Early work by Abstreiter et al. (2021) extended diffusion objectives to representation learning, showing that denoising naturally yields meaningful features. SODA (Hudson et al., 2024) augments a diffusion model with an information bottleneck to learn compact representations. Chen et al. (2024) deconstruct diffusion models and find that a plain denoising autoencoder suffices for strong self-supervised performance. Wang & He (2025) introduce a dispersive loss that encourages internal representation diversity in diffusion models, improving generation without external encoders, and Jiang et al. (2025) propose Self-Representation Alignment (SRA), which aligns a diffusion transformer’s latent features across noise levels to provide self-guidance without an auxiliary model. Collectively, these works establish denoising as a fundamental self-supervised task that induces robust features. Our findings complement this view: aligning diffusion models with strong spatial representations from self-supervised encoders improves both generative and discriminative capability, indicating that the preservation of spatial structure is central to this synergy.

Scaling self-supervised vision encoders. Recent years have seen substantial progress in scaling self-supervised vision models to unprecedented parameter counts and dataset sizes. DINOv3 (Siméoni et al., 2025) trains a 7B-parameter ViT on 1.7 billion unlabeled images by aligning representations across augmentations. WebSSL (Fan et al., 2025) shows that visual models trained on more than 2 billion images can match language-supervised models such as CLIP (Radford et al., 2021) on vision-language tasks without language supervision. C-RADIO (Heinrich et al., 2024) distills multiple teacher models into versatile encoders that generalize across visual domains. I-JEPA (Assran et al., 2023) studies predictive architectures that learn by predicting masked regions in representation space rather than reconstructing pixels, and SAM (Ravi et al., 2024) specializes in promptable segmentation via large-scale supervised training. The Perception Encoder family (Bolya et al., 2025; Cho et al., 2025) shows that intermediate features often outperform final representations on dense prediction. Although these models are typically benchmarked on global tasks such as image classification, our work identifies a key caveat: strong discriminative performance does not necessarily yield better generation. Instead, encoders that preserve spatial structure, irrespective of their classification accuracy, provide the greatest benefit for diffusion training. This argues for reconsidering the evaluation criteria for self-supervised encoders in generative applications, treating spatial coherence as no less important than semantic understanding.

I NOTE ON LLM USAGE

We use GPT-5 (OpenAI, 2025) to explore variations of the spatial structure metrics discussed in §3. All figures in the paper are generated directly from our experiment logs and checkpoints using Claude-Code (Anthropic, 2025). We additionally use LLM assistance to survey and formulate the related work in Appendix H, and we use Cursor in parts of the writing process.

J ADDITIONAL DISCUSSION ON SPATIAL STRUCTURE METRICS

Spatial structure metrics can quantify the representation gap. Yu et al. (2024) quantify the representation gap via linear probing, using the induced increase in validation accuracy to explain the effectiveness of REPA. We show that the spatial structure metrics (SSM) can likewise quantify this gap and account for the improvements across models. As shown in Figure 17, representation alignment narrows the gap between the SSM of a pretrained encoder such as DINOv2 and that of the diffusion features.

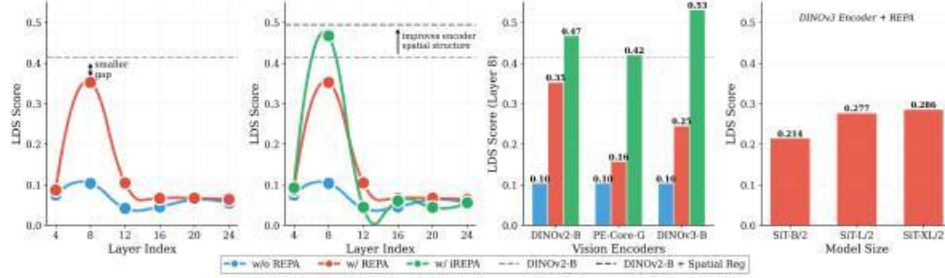


Figure 17: **Representation alignment as closing the spatial feature gap.** a) Representation alignment can be interpreted as closing the spatial feature gap between the diffusion features and the vision-encoder patch features. b) iREPA (§4) amplifies the spatial features of the vision encoder, at the cost of some global information, thereby improving generation. c) iREPA improves performance for every vision encoder considered. d) The spatial improvements scale with model size. All results are reported using SiT-XL/2 at 400K iterations.

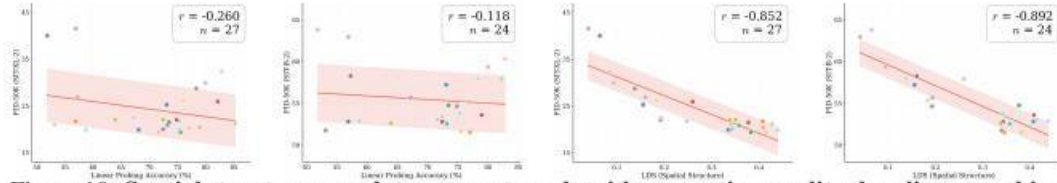


Figure 18: **Spatial structure correlates more strongly with generation quality than linear probing.** Correlation analysis over 27 vision encoders. **Left two:** Linear-probing accuracy against FID is weakly correlated, for SiT-XL-2 (Pearson $r = -0.26$) and SiT-B-2 ($r = -0.12$). **Right two:** Spatial structure (LDS) against FID is strongly correlated, for SiT-XL-2 ($r = -0.85$) and SiT-B-2 ($r = -0.89$). See Figure 10 in the Appendix for per-encoder labeled plots.

K ADDITIONAL ANECDOTAL COMPARISONS

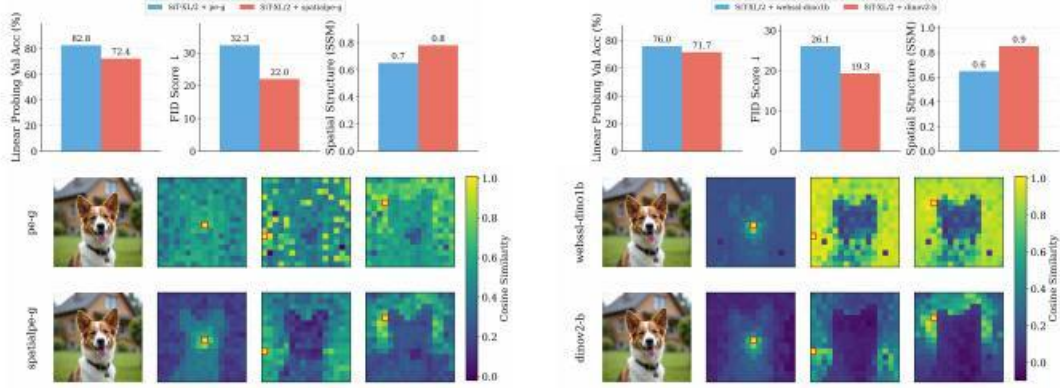


Figure 19: **Motivating anecdotes from recent SSL encoders — spatial structure matters.** **Top row:** Comparison of encoder metrics exhibiting an inverse relationship between ImageNet accuracy and generation quality. **Left:** PE-g attains higher accuracy (82.8% vs 72.4%) but worse FID (32.3 vs 22.0) than Spatial-PE-g, with substantially weaker spatial structure (LDS: 0.1 vs 0.4). **Right:** WebSSL-dino1b attains higher accuracy (76.0% vs 71.7%) but worse FID (26.1 vs 19.3) than DINOv2-b, again with weaker spatial structure (LDS: 0.2 vs 0.4). **Bottom row:** Spatial cosine-similarity visualizations confirm that the encoders with better generation (Spatial-PE-g, DINOv2-b) retain clear spatial coherence, whereas those optimized for classification (PE-g, WebSSL) lose spatial structure.

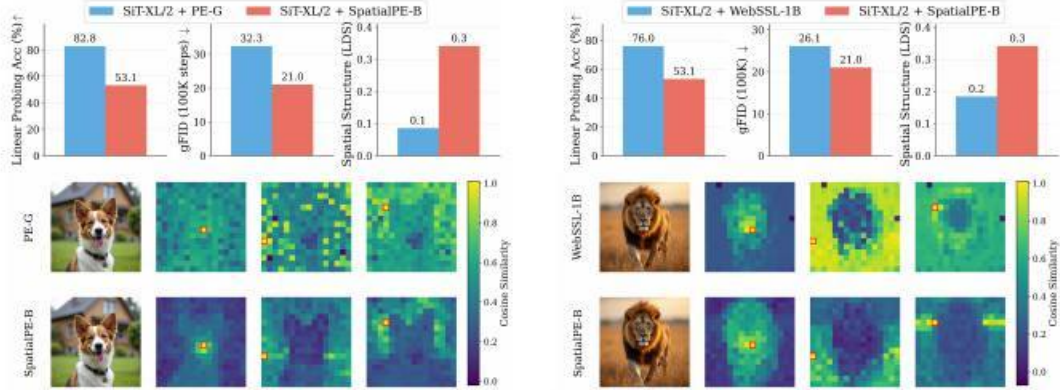


Figure 20: **Motivating examples — spatial structure matters.** **Top:** Comparison of encoder metrics exhibiting an inverse relationship between ImageNet accuracy and generation quality. **Left:** PE_{core-G}, despite substantially higher validation accuracy (82.8% vs. 53.1%), yields worse generation than PE_{Spatial-B} (Bolya et al., 2025). **Right:** Analogously, WebSSL-1B (Fan et al., 2025) attains substantially higher global performance (76.0% vs. 53.1%) yet worse generation. **Bottom:** Spatial structure provides a more reliable predictor of generation quality than global performance; see §3 for the spatial structure metric. All results reported at 100K iterations using SiT-XL/2 and REPA.

L FURTHER DISCUSSION

Role of Spatial Normalization We visualize the effect of spatial normalization on vision-encoder representations. Vision encoders can carry a substantial global component that suppresses the contrast among spatial tokens; for instance, tokens in one semantic region can be strongly correlated with tokens in unrelated regions (*e.g.*, background), reducing their spatial distinctiveness. Spatial normalization removes this global component, thereby increasing the contrast among spatial tokens. This allows the model to better preserve local spatial structure while attenuating the global information that would otherwise interfere with generation quality.

The following figures illustrate this effect across several examples, showing how spatial normalization reshapes the feature representations to emphasize spatial structure:

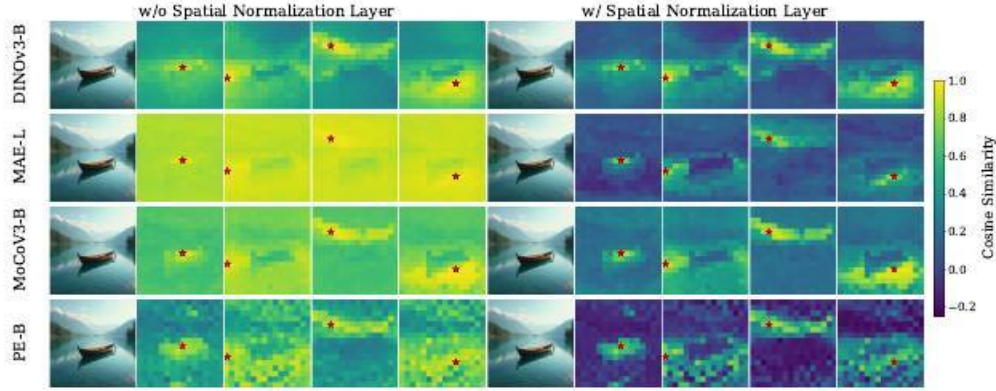


Figure 21: Visualizing the effect of spatial normalization (Example 1). The heatmaps show token-similarity patterns before and after spatial normalization. Without normalization (left), the global component induces high correlations across unrelated regions; with spatial normalization (right), local spatial structure is enhanced and global interference is suppressed, yielding more distinct semantic boundaries.

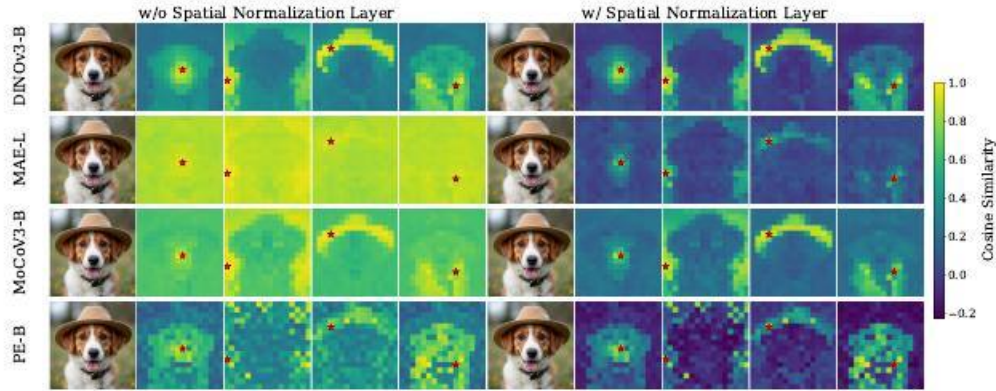


Figure 22: Visualizing the effect of spatial normalization (Example 3). The heatmaps show token-similarity patterns before and after spatial normalization. Without normalization (left), the global component induces high correlations across unrelated regions; with spatial normalization (right), local spatial structure is enhanced and global interference is suppressed, yielding more distinct semantic boundaries.

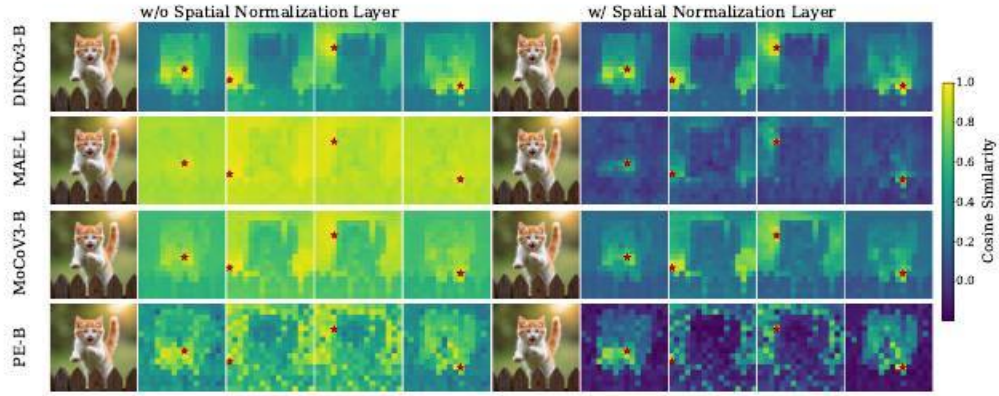


Figure 23: Visualizing the effect of spatial normalization (Example 2). The feature-similarity maps illustrate how spatial normalization increases spatial contrast. The original features (left) carry a global overlay that reduces the distinction between foreground and background; after normalization (right), the spatial tokens become more locally coherent, with clearer separation between semantic regions.

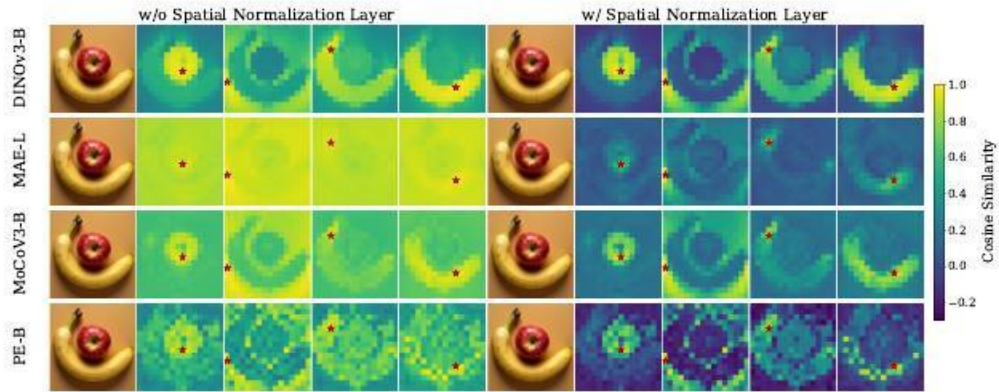


Figure 24: **Visualizing the effect of spatial normalization** (Example 4). The heatmaps show token-similarity patterns before and after spatial normalization. Without normalization (left), the global component induces high correlations across unrelated regions; with spatial normalization (right), local spatial structure is enhanced and global interference is suppressed, yielding more distinct semantic boundaries.

M ADDITIONAL RESULTS

M.1 ADDITIONAL RESULTS AT HIGHER RESOLUTIONS

Setup. We conduct further experiments on ImageNet-512 (Deng et al., 2009) to evaluate whether the proposed spatial modifications generalize to higher resolutions. Following the REPA (Yu et al., 2024) setup, we report results across a range of pretrained encoders (DINOv2, DINOv3, WebSSL, PE, etc.), using SiT-XL and SiT-B with 50 NFE at inference and without classifier-free guidance. Fig. 25 reports REPA generation before and after applying the spatial modifications (iREPA). The modifications generalize to higher resolutions, yielding consistent gains across all considered pretrained encoders (DINOv2, DINOv3, WebSSL, PE, etc.).

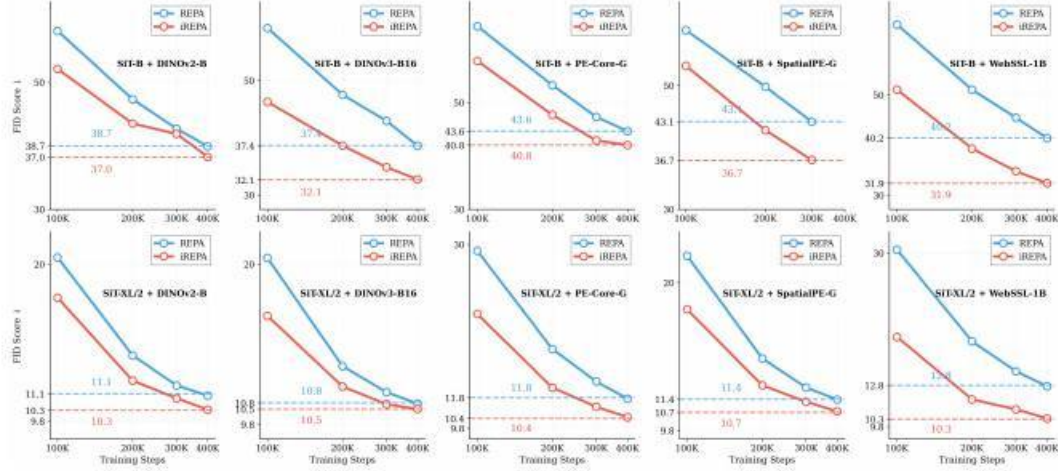


Figure 25: **Convergence results at 512×512 resolution.** Amplifying spatial features consistently accelerates convergence across resolutions, for both ImageNet-256 (Figure 7) and ImageNet-512 (above).

M.2 ADDITIONAL RESULTS ON MULTIMODAL T2I TASKS

Setup. To assess the generalization of the proposed spatial modifications beyond ImageNet, we conduct extensive experiments on multimodal text-to-image (T2I) tasks. Following REPA (Yu et al., 2024), we adopt MMDiT (Esser et al., 2024) as the diffusion backbone and apply REPA with a range of pretrained vision encoders (DINOv2, CLIP, WebSSL, PE, etc.). As in REPA (Yu et al., 2024), all models are trained for 150K steps with batch size 256 and evaluated with an ODE sampler using 50 NFE. Fig. 26 reports REPA generation before and after applying the spatial modifications (iREPA). The modifications generalize to multimodal T2I tasks, yielding consistent gains across all considered pretrained encoders (DINOv2, CLIP, WebSSL, PE, etc.).

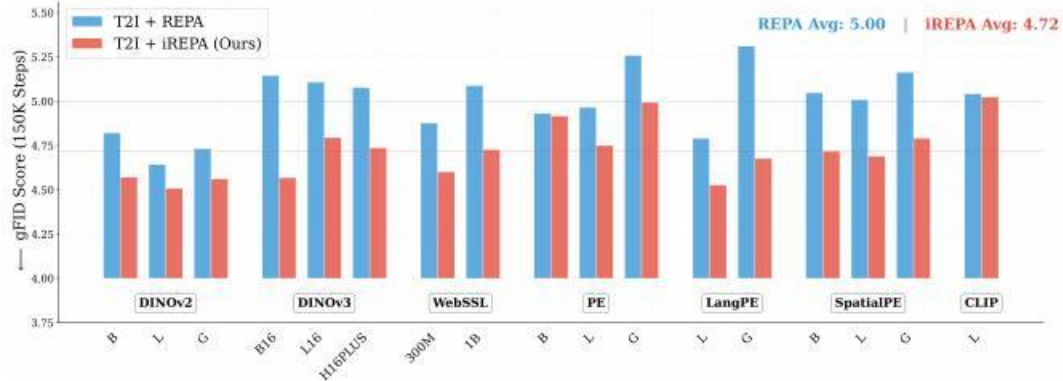


Figure 26: **Text-to-image generation across encoder variants.** Amplifying the transfer of spatial features from the target representation to the diffusion features consistently accelerates convergence for both ImageNet (see §4) and multimodal T2I tasks (above), with consistent gains across all considered pretrained encoders (DINOv2, CLIP, WebSSL, PE, etc.).

M.3 CORRELATION ANALYSIS WITHOUT OUTLIERS (MoCOV3-L AND MAE-L)

We repeat the correlation analysis of Section 3 after excluding MoCOV3-L and MAE-L, which are identified as outliers. Figure 28 reports the updated correlations across model sizes. Spatial structure remains far more strongly correlated with generation performance than linear-probing accuracy. Notably, once the outliers are removed, linear probing exhibits a small *positive* correlation with gFID, i.e., higher linear-probing accuracy is associated with worse generation. This is consistent with the observations of Sec. 2, where target representations with higher global semantic performance (linear-probing accuracy) frequently yield comparable or worse generation under representation alignment (REPA).

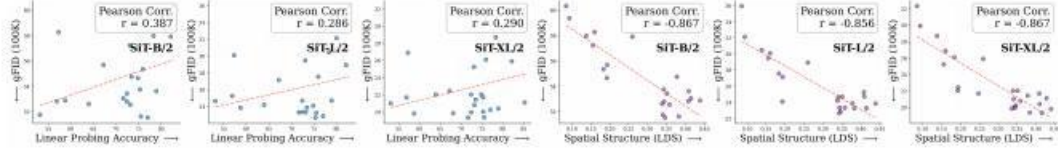


Figure 27: Correlation analysis without outliers. After removing the outliers (MoCOV3-L and MAE-L), spatial structure remains far more strongly correlated with generation performance (Pearson $|r| > 0.85$) than linear-probing accuracy (Pearson $|r| < 0.38$) at every model size (B, L, XL). Notably, with the outliers removed, linear probing exhibits a small positive correlation with gFID (higher linear-probing accuracy is associated with worse generation). This is consistent with the observations of Sec. 2, where target representations with higher global semantic performance (linear-probing accuracy) frequently yield comparable or worse generation under representation alignment (REPA).

M.4 ADDITIONAL QUALITATIVE RESULTS



Figure 28: **Additional Qualitative Results** comparing generation outputs before and after applying the spatial modifications to REPA. All results use PE-G (Bolya et al., 2025) as the pretrained vision encoder, 400K steps (80 epochs), and a classifier-free guidance scale of 4.0. Consistent with the quantitative gains (§4), the spatial modifications (iREPA) also improve the visual quality and coherence of the generated outputs.

M.5 ADDITIONAL RESULTS WITH ALTERNATIVE EVALUATION METRICS

Beyond the conventional evaluation metrics (Inception Score, FID, sFID, Precision, Recall), we verify the robustness of our findings under alternative measures of generation quality (Stein et al., 2023; Yang et al., 2023; Kynkäänniemi et al., 2023; Jayasumana et al., 2024). Specifically, we adopt the CMMD metric (Jayasumana et al., 2024) with the PyTorch implementation of (Paul, 2024), using the standard OpenAI ADM evaluation suite (Dhariwal & Nichol, 2021) for the reference images. We then re-verify the principal findings of §3 and §4 under CMMD.

Spatial structure correlates far more strongly with generation performance than linear probing.

To test the robustness of the analysis in §3, we repeat the large-scale correlation analysis across vision encoders using the CMMD metric (Jayasumana et al., 2024). Results are reported in Figure 29, using SiT-B/2 (100K steps) and REPA. Every spatial metric attains a substantially stronger correlation (Pearson $|r| > 0.88$) with generation performance (CMMD) than linear probing (Pearson $|r| = 0.074$), confirming that the key findings of §3 are invariant to the choice of evaluation metric (FID or CMMD).

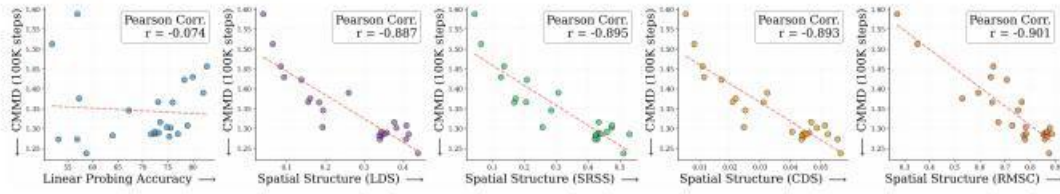


Figure 29: **Correlation analysis under the CMMD metric.** We repeat the correlation analysis of §3 using CMMD (Jayasumana et al., 2024) in place of gFID. All results use SiT-B/2 (100K steps) and REPA. Every spatial metric attains a substantially stronger correlation (Pearson $|r| > 0.88$) with generation performance (CMMD) than linear probing (Pearson $|r| = 0.074$), confirming that the key findings of §3 are invariant to the choice of evaluation metric.

Amplifying the transfer of spatial features improves generation quality. We next test the generalization of the analysis in §4 under CMMD (Jayasumana et al., 2024), in addition to the conventional metrics (gFID, sFID, IS, etc.). Results are reported in Table 8. Across all vision encoders, amplifying the transfer of spatial features (iREPA) accelerates convergence under both CMMD and the conventional metrics (IS, FID, sFID, Prec., Recall), with consistent improvements both with and without classifier-free guidance (CFG).

Vision Encoder	Steps	w/o CFG						w/ CFG					
		CMMD↓	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑	CMMD↓	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑
DINOv2-B	100K	0.702	69.20	19.3	5.89	0.64	0.61	0.529	157.2	6.35	5.91	0.77	0.54
+iREPA	100K	0.652	77.92	16.9	6.26	0.66	0.61	0.484	179.3	5.15	6.23	0.78	0.54
DINOv2-B	400K	0.455	127.4	7.76	5.06	0.70	0.66	0.320	263.0	1.98	4.60	0.80	0.61
+iREPA	400K	0.438	128.6	7.52	4.89	0.71	0.65	0.310	268.8	1.93	4.59	0.80	0.60
DINOv3-B	100K	0.749	63.64	21.4	6.14	0.63	0.60	0.571	144.0	7.57	6.09	0.76	0.53
+iREPA	100K	0.651	78.79	16.2	6.14	0.66	0.61	0.481	181.9	4.87	6.10	0.78	0.55
DINOv3-B	400K	0.474	126.7	8.10	5.06	0.70	0.66	0.336	261.2	1.99	4.58	0.80	0.61
+iREPA	400K	0.441	132.9	7.13	4.93	0.71	0.66	0.314	272.4	1.89	4.58	0.80	0.60
WebSSL-1B	100K	0.825	53.87	25.5	6.57	0.61	0.59	0.622	124.0	9.59	6.37	0.76	0.52
+iREPA	100K	0.653	77.47	16.6	6.18	0.66	0.61	0.484	177.6	5.09	6.11	0.79	0.54
WebSSL-1B	400K	0.512	116.9	9.39	5.14	0.70	0.64	0.354	250.5	2.24	4.61	0.81	0.58
+iREPA	400K	0.445	130.8	7.48	4.91	0.70	0.65	0.311	271.6	1.90	4.58	0.80	0.61
PE-Core-G	100K	0.922	42.74	32.3	6.70	0.57	0.59	0.714	97.2	14.1	6.56	0.71	0.53
+iREPA	100K	0.697	75.01	18.1	6.03	0.64	0.61	0.525	176.8	5.66	6.08	0.77	0.54
PE-Core-G	400K	0.525	109.4	10.4	5.00	0.69	0.64	0.366	238.4	2.44	4.57	0.81	0.59
+iREPA	400K	0.458	132.0	7.78	5.02	0.70	0.65	0.322	275.4	1.93	4.59	0.80	0.61
CLIP-L	100K	0.790	54.07	25.2	6.65	0.61	0.60	0.605	124.5	9.77	6.59	0.75	0.52
+iREPA	100K	0.657	74.46	17.8	6.33	0.65	0.61	0.489	172.0	5.67	6.33	0.78	0.54
CLIP-L	400K	0.487	117.8	8.97	4.98	0.70	0.65	0.339	258.7	2.15	4.64	0.81	0.59
+iREPA	400K	0.442	130.1	7.42	5.03	0.71	0.65	0.307	271.8	1.96	4.64	0.80	0.61

Table 8: **Additional results under alternative evaluation metrics.** We report supplementary results using the CMMD metric (Jayasumana et al., 2024). All results are obtained using SiT-XL/2 with batch size 256 and standard REPA as the baseline. CMMD is computed with the PyTorch implementation of (Paul, 2024), and the reference images are drawn from the standard OpenAI ADM evaluation suite (Dhariwal & Nichol, 2021). Across all vision encoders, amplifying the transfer of spatial features (iREPA) accelerates convergence under both CMMD and the conventional metrics (IS, FID, etc.), confirming that the key findings of §4 are invariant to the choice of evaluation metric.

M.6 SPATIAL IMPROVEMENTS WITH SAM2

Table 9 and Figure 30 examine the effect of spatial normalization (§4) on the SAM2-S (46M) encoder features². As Table 9 shows, the spatial normalization layer improves SAM2 performance, but only marginally. The explanation is structural: spatial normalization enhances spatial contrast by removing the global component (the mean of the patch tokens). Because SAM2 already carries little to no global information (validation accuracy < 24%), this removal produces only a small increase in spatial contrast (see Figure 30).

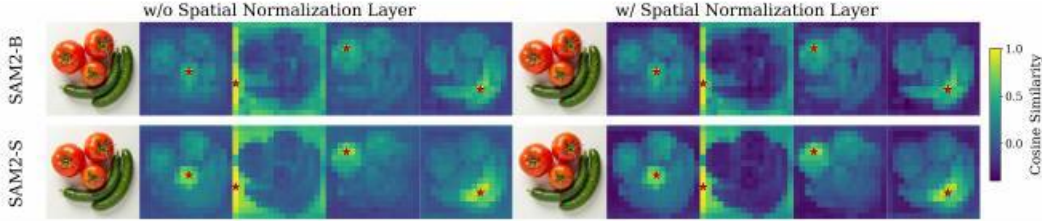


Figure 30: **Effect of spatial normalization on SAM2.** For SAM2, the spatial normalization layer enhances spatial contrast only marginally. The reason is structural: spatial normalization (§4) enhances spatial contrast by removing the global component (the mean of the patch tokens), and SAM2 already carries little to no global information (validation accuracy < 24%), leaving little to remove.

Vision Encoder	Steps	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
SAM2-S	100K	50.69	25.32	6.52	0.631	0.588
+iREPA	100K	53.14	24.52	6.28	0.629	0.591
SAM2-S	400K	110.77	9.54	4.93	0.699	0.638
+iREPA	400K	114.62	9.10	4.89	0.704	0.640

Table 9: **Effect of the spatial modifications on SAM2.** All results use SiT-XL/2 without classifier-free guidance, SAM2-S (46M) as the vision encoder, and standard REPA as the baseline.

²As in (Bolya et al., 2025), we use the intermediate vision-encoder output for the SAM2 features rather than the mask logits. As shown in (Bolya et al., 2025), the mask logits yield sharper spatial maps but are not themselves suitable as a target representation.

M.7 ADDITIONAL RESULTS WITH FULL-FINETUNING ACCURACY INSTEAD OF LINEAR PROBING ACCURACY

Correlation analysis using full-finetuning accuracy in place of linear-probing accuracy. We repeat the correlation analysis of §3 using post-full-finetuning validation accuracy³ rather than linear probing. Results are reported in Fig. 31, using SiT-XL/2 (100K steps) and REPA. For the full-finetuning accuracy we finetune the entire vision encoder with a learning rate of $5e-5$, a warmup ratio of 0.1, and 30 epochs. Every spatial metric attains a substantially stronger correlation with generation performance than full-finetuning accuracy. Notably, gFID exhibits a weak positive correlation with post-full-finetuning validation accuracy (Pearson $r = 0.317$): higher validation accuracy is associated with higher gFID, i.e., worse generation. This is consistent with the observations of §2, where target representations with higher global semantic performance frequently yield comparable or worse generation under representation alignment (REPA).

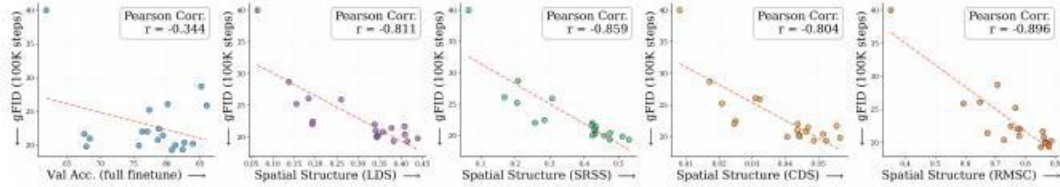


Figure 31: Correlation analysis using full-finetuning accuracy in place of linear-probing accuracy. We repeat the correlation analysis of §3 using post-full-finetuning validation accuracy rather than linear probing. All results use SiT-XL/2 (100K steps) and REPA. Every spatial metric attains a substantially stronger correlation with generation performance than full-finetuning accuracy. Notably, gFID exhibits a weak positive correlation with post-full-finetuning validation accuracy (Pearson $r = 0.317$): higher validation accuracy is associated with higher gFID, i.e., worse generation. This is consistent with the observations of §2, where target representations with higher global semantic performance frequently yield comparable or worse generation under representation alignment (REPA).

³While the findings are qualitatively unchanged, linear probing is arguably the more faithful estimator of global information for REPA. REPA regularizes with the *pretrained* encoder features themselves, whereas finetuning the encoder alters the amount of global information it carries. Following REPA (Yu et al., 2024), we therefore adopt linear-probing accuracy as the default measure of global information.

M.8 ADDITIONAL RESULTS ON PIXEL-SPACE DIFFUSION MODELS

Method	#Params	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
PixelFlow (Chen et al., 2025)	459M	24.67	54.33	9.71	-	0.58
PixDDT (Wang et al., 2025b)	434M	36.24	46.37	17.14	-	0.63
PixNerd (Wang et al., 2025a)	458M	43.01	37.49	10.65	-	0.62
DeCo w/o $\mathcal{L}_{\text{FreqFM}}$	426M	46.44	34.12	10.41	-	0.64
DeCo + REPA (Ma et al., 2025b)	426M	48.35	31.35	9.34	-	0.65
JiT (Li & He, 2025a)	459M	29.37	49.06	11.21	0.40	0.62
JiT+iREPA (Ours)	459M	50.72	29.19	9.42	0.51	0.65

Table 10: **Generation performance of pixel-space diffusion models.** Notwithstanding its simplicity, iREPA combined with JiT (Li & He, 2025a) outperforms recent state-of-the-art pixel-space diffusion methods such as DeCo (Ma et al., 2025b). All results are obtained at 200K training iterations with batch size 256 and evaluated using 50-step Euler sampling without classifier-free guidance. Results for DeCo are taken directly from (Ma et al., 2025b).